# AI brainstorming: Is less context more?

Narrow context is more efficient than full context

for generative ideation with product graphs*

Almubdi Muitakan[1] Maria Tsakalerou[2] Marc Magrans de Abril[3] Jose Berengueres[1]

[3]Independent researcher
[1]School of Computing and AI, Nazarbayev University
[2]School of Engineering, Nazarbayev University

**Abstract**

Instead of ephemeral human-vs-AI benchmarks, we evaluate representation efficiency: standout ideas per total token spent. Using a 6-layer microwave oven knowledge graph (123 nodes, 246 edges), we test four prompt representations across 13 market needs and four thinking-effort levels: full graph (A), market-need vertical slice (B), size-matched random slice (R, controlling for prompt length), and no graph (C). Claude 3.5 Haiku generated 2,892 ideas (580 calls) under the SCAMPER Substitute operator, scored blind by Claude 3.5 Sonnet.

We measure yield: top-decile ideas (top 12%, score $\geq 4.0/5.0$) per 100k total tokens (input plus output/reasoning). The targeted market-need slice at medium effort achieves peak yield (8.8 top ideas/100k tokens)—nearly triple the full graph and superior on 10/13 market needs. While the full graph produces more detailed average sets, it yields no more top ideas and inflates token cost. An additional step of human expert analysis also shows that, despite high scores, top ideas retrieve established state-of-the-art solutions rather than unpatented inventions: LLM ideation operates as query-driven retrieval, where narrow, structured context maximizes discovery throughput. Thinking longer raised linguistic quality metrics but not token efficiency.

## 1 Introduction

Most research on AI for innovation asks about outcomes: whether LLM-generated ideas rival human ones (Si et al., 2024; Terwiesch et al., 2026), whether AI assistance elevates individual or team creativity (Doshi and Hauser, 2024; Anderson et al., 2024), and whether agentic pipelines outperform single prompts or human designers (Liu et al., 2026; Massoudi and Fuge, 2026). Yet outcome benchmarks against human baselines expire rapidly as model capabilities expand and token prices fall. A more enduring question is how to **present the problem** to the model: which representation of a product allows generative AI to turn compute into standout innovation concepts most efficiently.

In product ideation, engineering design, and service innovation, problem representations range from business model canvases and user journey maps to design structure matrices and typed knowledge graphs (Akao, 2024; Steward, 1981; Gero, 1990). Here, we focus on the *innovation matrix* (IM) of Leo Tchirsky (Tchirsky, n.d.; Berengueres, 2014)—a layered, typed

*Code, data, prompts and all model outputs: `https://github.com/orioli/IM-LLM`

graph connecting market needs to functions, components, enabling technologies, and scientific fields. While cost-aware evaluation is now standard for LLM reasoning and agent benchmarks (Chen et al., 2024a; Wang et al., 2024; Kapoor et al., 2025), generative ideation is still largely judged by average scores without accounting for token expenditure or the extreme-value nature of brainstorming (Girotra et al., 2010; Dahan and Mendelson, 2001).

To address this gap, we examine how problem representations and reasoning allocations govern AI brainstorming efficiency across three questions: first, which graph representation yields the most top ideas per token (RQ1); second, whether structural relevance or raw prompt length drives performance (RQ2); and third, how investing tokens in context compares with spending them on thinking effort (RQ3). We test these questions using a countertop microwave oven represented as an innovation matrix with 123 nodes and 246 edges. Using Claude 3.5 Haiku as the ideation model and the *Substitute* operator from the SCAMPER ideation framework (Eberle, 1971; Osborn, 1953) across 13 market needs, we evaluate four prompt representations: the full graph (A), a targeted vertical slice containing only direct child nodes of the active market need (B), a size-matched random slice (R), and a no-graph control (C). Each condition is crossed with four thinking-effort levels across 580 API calls, generating 2,892 ideas that are evaluated blind by Claude 3.5 Sonnet and validated by human expert review. We measure *yield*: standout ideas (top $\sim$12%, score $\geq 4.0/5.0$) per 100k total tokens spent.

Our primary finding is that prompting with a targeted market-need slice at medium effort achieves peak yield (8.8 top ideas per 100k tokens), nearly tripling the efficiency of the full graph. Although the full graph produces more specific average sets, it yields no more top-tail ideas while doubling prompt costs. Furthermore, scaling thinking effort raises average linguistic quality but fails to improve token efficiency. Together, these findings make three contributions: they reframe AI ideation around representation efficiency (outcome per token) rather than ephemeral human baselines; they establish yield as a non-saturating tail metric that captures extreme-value creative discovery where mean scores mislead; and they provide an open, controlled empirical comparison of product-graph representations, thinking effort, and prompt personas, with all prompts, calls, and judgments released.

The remainder of this paper is structured as follows. Section 2 reviews related work on design representations, LLM ideation workflows, and cost-aware evaluation. Section 3 details the product graph, prompt representations, experimental design, and scoring metrics. Section 4 presents empirical findings for RQ1–RQ3, exploratory persona analyses, and robustness checks. Finally, Section 5 discusses implications for design thinking and outlines directions for graph-guided AI innovation.

# 2 Related work

## 2.1 Innovation Matrix background

Structured descriptions that link what customers want to what a product does and what it is made of are older than computing in design. Quality function deployment began in Japanese manufacturing in the late 1960s, was taken up by Toyota and its suppliers in the 1970s, and reached Western industry as the "house of quality" (Akao, 2024; Hauser and Clausing, 1988; Chan and Wu, 2002). Its matrices relate customer requirements to engineering characteristics, and then to parts and processes, so that every design decision can be traced to a market need. Related schemes followed, each making a different link explicit: morphological analysis lays out the alternatives for each function (Zwicky, 1967; Ritchey, 2006, 2018); the design structure matrix records the dependencies between components (Steward, 1981; Browning, 2001; Eppinger and Browning, 2012); function–behaviour–structure separates what a design is for from what it is (Gero, 1990; Gero and Kannengiesser, 2004); the functional basis standardises the vocabulary of functions and flows (Stone and Wood, 2000; Hirtz et al., 2002); and technology roadmaps link market needs to products and technologies over time (Phaal et al., 2004). The innovation matrix

used here (Tchirsky, n.d.; Berengueres, 2014) combines several of these links in one layered graph, from market needs through functions and components to technologies and scientific fields (it is distinct from the design-led innovation matrix of Kyffin and Gardien (2009)). All of these were made for people, typically a team in front of a wall chart, who read them selectively.

### 2.2 LLMs that innovate with design methods

Language models were first used as concept generators (Zhu and Luo, 2022; Ma et al., 2023) and compared with people: their concepts are less diverse than crowdsourced ones (Ma et al., 2025), their research ideas are judged more novel but slightly less feasible than experts' (Si et al., 2024), and working with them raises individual creativity while narrowing collective diversity (Doshi and Hauser, 2024; Anderson et al., 2024; Ashkinaze et al., 2025; Wadinambiarachchi et al., 2024). Recent systems give the model the structured methods above. An FBS ontology steers concept generation (Chen et al., 2024e); generated text and images enrich concepts (Chen et al., 2024b); TRIZ is applied by single models and by agent teams (Chen et al., 2024c; Jiang et al., 2025; Szczepanik and Chudziak, 2025); morphological analysis is automated (Chen et al., 2024d); and agentic workflows chain whole design processes (Ding et al., 2023; Massoudi and Fuge, 2026). iDesignGPT (Liu et al., 2026) goes furthest: agent clusters run problem definition with QFD and Kano analysis, then functional decomposition, morphological analysis, SCAMPER (Eberle, 1971) and TRIZ, and produce a product design specification. Persona prompts are a related device, from role-play that raises the creativity of LLM discussions (Lu et al., 2024) to agents that act out a design team (Liu et al., 2025), drawing on ideas about thinking styles in innovation (de Bono, 1970; Kelley and Littman, 2005).

### 2.3 What "innovative" is checked against

These systems claim to innovate, but the claim usually rests on ratings: experts or a model score novelty and originality, or embedding distances measure how far the generated concepts spread from each other. iDesignGPT, for example, reports that it "surpasses benchmark models in generating innovative, modular, and rational solutions", on the basis of expert scores against GPT-4o baselines and semantic novelty metrics; patents are searched as an input to the workflow, but the generated concepts are not compared with prior art (Liu et al., 2026). This is typical. Novelty is judged relative to other generated ideas or to a rater's memory, seldom against what already exists, although patent-based measures of technological novelty are well established (Fleming, 2001; Verhoeven et al., 2016; Uzzi et al., 2013) and patent-derived knowledge networks are available for design (Sarica et al., 2020; Luo et al., 2021). An idea a rater finds novel may be on the market already. Our own study shares this limitation: its judge rates novelty and patentability from knowledge, without a prior-art search, and several of its top-rated ideas, such as solid-state RF heating and RF thawing, already exist in industry, as the judge itself noted. A second gap is cost: none of these systems reports what its outputs cost, so a more elaborate pipeline cannot be weighed against a simpler one. None of these experiments has resulted to our knowledge in a single patent. Perhaps in our opinion due to token costs of implementing genchibembutsu Berengueres (2014), collecting first hand information, and collecting field information.

### 2.4 Context and cost in LLMs

More context is not free even when it fits in the window. In practice, models exhibit an "effective context window" far smaller than their advertised capacity, with performance degrading sharply as input token length increases (Paulsen, 2026), a limitation driven by positional frequency biases and attention sparsity that restrict effective context scaling (An et al., 2024). Models use information in the middle of long inputs poorly (Liu et al., 2024), their reasoning degrades

as inputs grow long well before the limit (Levy et al., 2024), and they struggle when relevant content must be inferred rather than matched (Modarressi et al., 2025); retrieval can reach results comparable to long context at a fraction of the cost (Li et al., 2024). Elsewhere, cost is becoming a first-class outcome: cascades and routers match strong models at lower cost (Chen et al., 2024a; Ong et al., 2024; Ding et al., 2024), reasoning strategies are compared at equal token budgets (Wang et al., 2024; Han et al., 2025), and agent benchmarks report accuracy together with cost (Kapoor et al., 2025; Wang et al., 2025). This concerns tasks with defined right answers. Ideation, as mentioned, has no defined right answers. Its efficiency is rarely considered.

### 2.5 Measuring ideation

For lack of objective measures (beyond patentability), currently ideation with AI is using linguistic-like tools. It is assessed by quantity, quality, (linguistic) novelty and (linguistic) variety (Shah et al., 2003), or by novelty and usefulness together (Sarkar and Chakrabarti, 2011). But a brainstorm is worth its best ideas, since the rest are discarded: idea-generation processes should be compared on the quality of the best idea, not the average (Girotra et al., 2010), and extreme-value models describe how the best of many concepts improves with the number tested (Dahan and Mendelson, 2001). Mean scores, the usual summary in LLM ideation studies, hide exactly this difference. To avoid this trap, here we combine the tail and the cost in one measure, top ideas per token, and vary only how the product is presented, holding the model, the method and the prompt fixed.

## 3 Methods

The experiment holds the AI model and the prompt fixed, and varies only how much of the product graph the prompt contains (the *representation*) and how long the model may think (the *effort*). It measures what each combination produces and what it costs.

### 3.1 Product graph

**The innovation matrix.** The innovation matrix (IM) of Leo Tchirsky (Tchirsky, n.d.), as taught in design-thinking workshops (Berengueres, 2014), describes a product as stacked layers: the **market needs** it serves, the **functions** that serve them, the **components** that perform the functions, the **technologies** that make the components possible, and the scientific **fields** behind those technologies (Figure 1). Reading upwards answers *why* a part exists; reading downwards answers *how* a market need is met. Innovation happens by changing a link: a new technology enables a different component, a component takes over another's function, or a function is found to serve a market need nobody had linked it to. In a workshop the matrix is drawn on a wall; here it is a typed graph that a model can read.

**The microwave matrix.** The product is a countertop microwave oven, the simplest mechanical model: a timer knob, a power knob, a door button and an interior lamp. The authors built its IM by hand in a single YAML file (`microwave_base_graph.yaml`), adding two cross-cutting types the workshop matrix keeps beside it, constraints and actors. It has 123 nodes and 246 directed, typed edges (Figure 2; all nodes listed in Appendix A).

**Nodes** belong to four stacked layers and four cross-cutting types (Table 1):

Table 1: Node types in the microwave innovation matrix (123 nodes): count per type and what each type represents.

| Type | Count | Role |
|---|---|---|
| MarketNeed (M) | 17 | what users want (e.g. M4 *Defrost frozen food quickly*) |

Table 1 (continued)

| Type | Count | Role |
|---|---|---|
| Function (F) | 20 | what the product does |
| Component (C) | 28 | physical parts |
| Technology (T) | 24 | the principles that enable components and functions |
| Field (D) | 10 | scientific fields the technologies are rooted in |
| Constraint (X) | 15 | limitations and conflicts (e.g. X3 *uneven heating*) |
| Actor (A) | 8 | user groups (e.g. A5 *parent with young children*) |
| Product (P) | 1 | the microwave oven |

**Edges** carry one of twelve types. Those that build the layer stack are `has_component`, `performs_function`, `enables_function`, `addresses_need`, `enabled_by` and `rooted_in`. Component-level flows are `feeds` and `actuates`. Constraint links are `causes`, `contradicts` and `partially_mitigates`. Weak associations are `relevant_to`. A thirteenth type, `resolves`, is declared in the schema for agents to use when a new node replaces or solves an existing one; it does not occur in the base graph.

Each node has an identifier, type, label and a one-line detail. The file begins with a meta block that names the layers and defines the edge types.

### 3.2 Representations

Figure 3 shows what the model sees in each condition. Every prompt contains a graph file produced by one serializer from the base graph. Fields, ordering and the meta block are the same in every file; only *which* nodes and edges are present differs. No file states which condition it belongs to. For each target market need $M_k$ (`slicer.py`):

- **A, full graph.** All 123 nodes and 246 edges. One file shared by all market needs.
- **B, market-need slice.** The vertical slice of $M_k$, obtained by walking down the matrix from the market need:

  1. the functions that `addresses_need` $M_k$, plus every function that enables them through `enables_function`, recursively;
  2. the components that `performs_function` one of those functions;
  3. the technologies that enable those components or functions (`enabled_by`);
  4. the fields those technologies are `rooted_in`;
  5. every constraint linked to any node above by `causes`, `contradicts`, `partially_mitigates` or `relevant_to`, and every actor `relevant_to` $M_k$;
  6. the product node.

  The file contains the edges induced among the selected nodes.
- **R, random slice.** The target market need and the product, plus a random draw of nodes with the same number of nodes of each type as B, with the induced edges. The draw is seeded by the market need identifier (`random.Random("<need>-1")`), so it is reproducible and independent of run order. R controls for the *size* and layer composition of B while removing its *relevance* to the market need.
- **C, no graph.** The meta block (layer names and edge-type definitions) and the target market need node only.

Four needs (M2, M9, M10, M15) are not addressed by any function, so their vertical slice is empty. They were excluded, leaving 13 target market needs: M1, M3, M4, M5, M6, M7, M8, M11, M12, M13, M14, M16, M17. Table 2 gives the size of each representation.

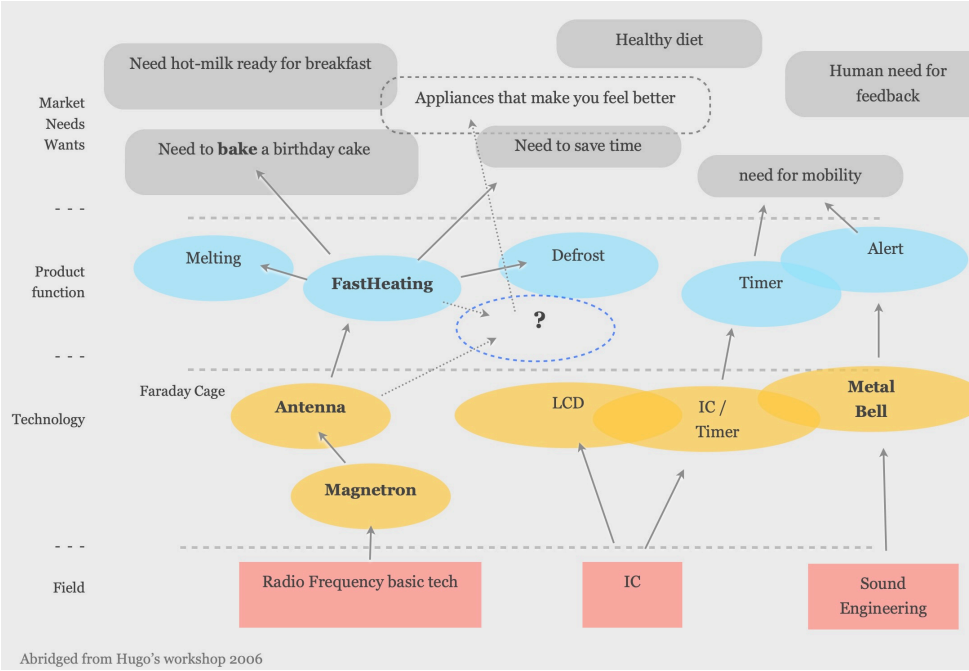



Figure 1: An innovation matrix as drawn in the workshop material (Berengueres, 2014).

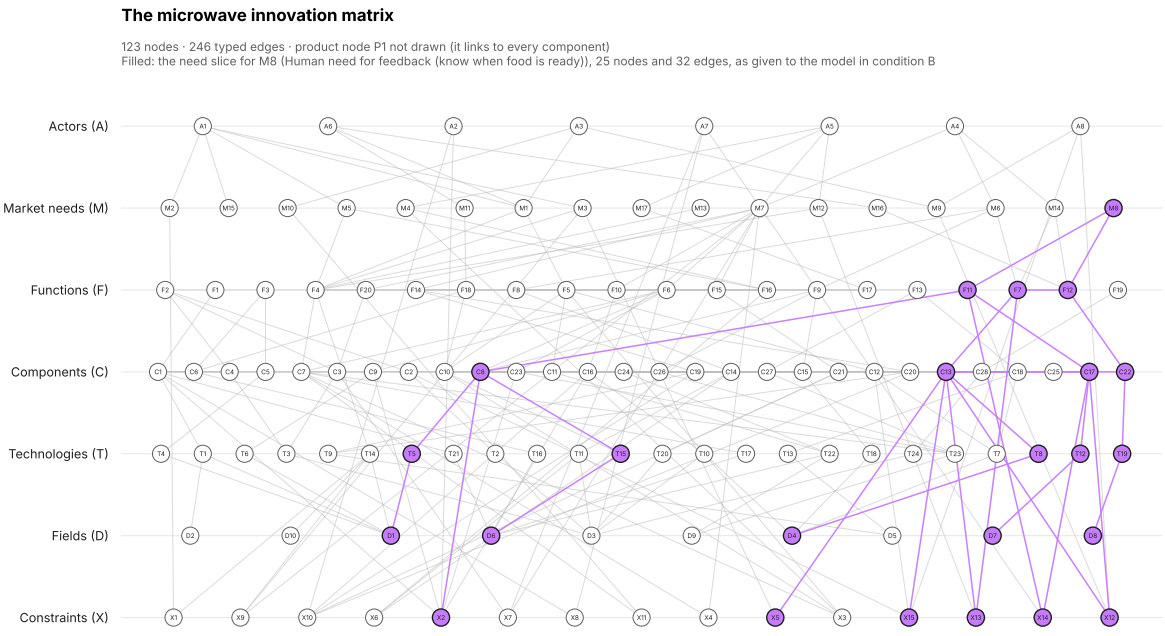



Figure 2: A slice of the graph based on market need M8 is shown highlighted in purple. The microwave innovation matrix used in the experiment. Rows are the layers, bottom to top, with constraints below and actors above. Filled nodes and edges are the market-need slice for M8, *know when food is ready* (25 nodes, 32 edges): what the model sees in condition B for that market need, against the whole graph in condition A.

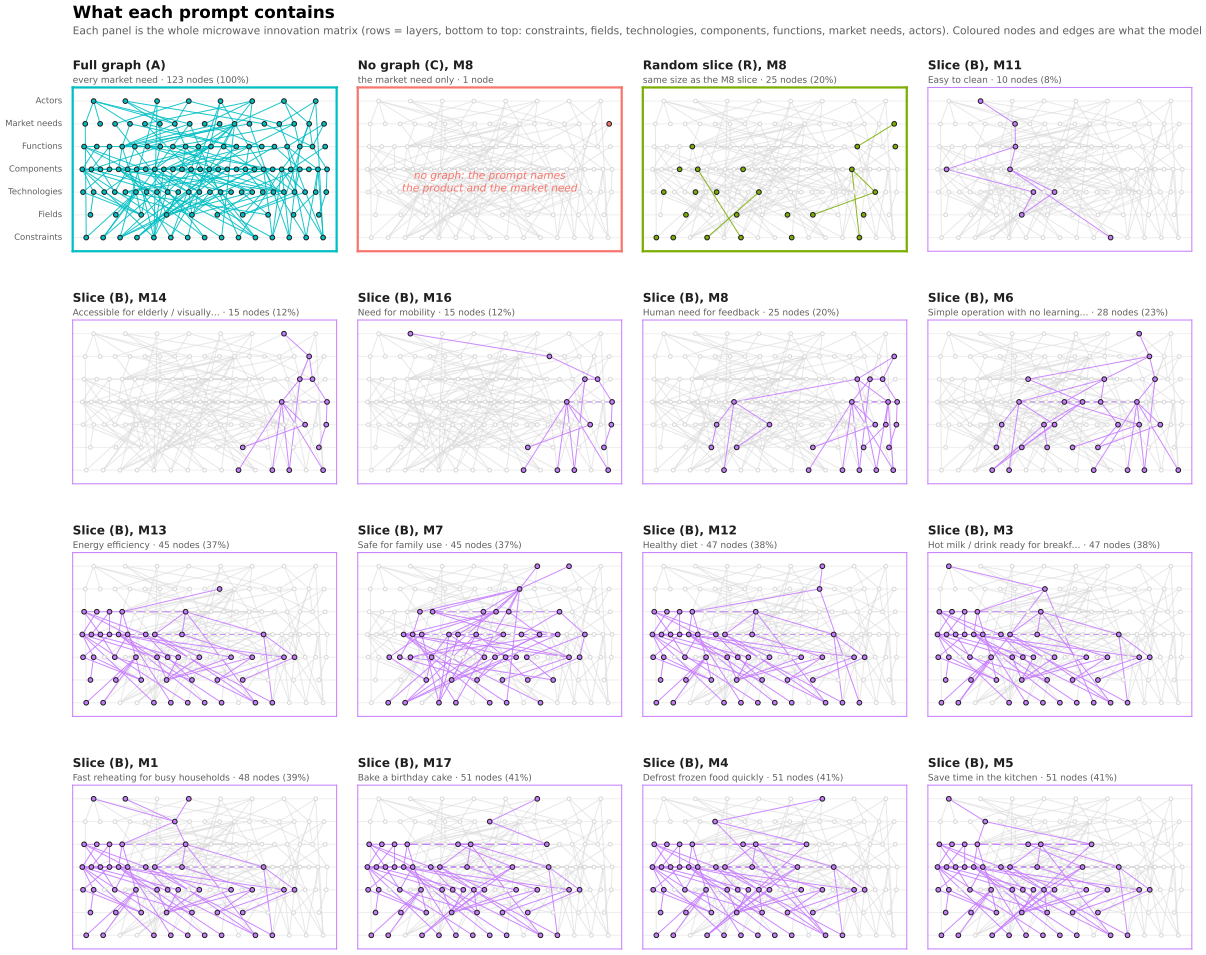



Figure 3: What each prompt contains. Every panel is the whole microwave innovation matrix (rows are the layers, bottom to top: constraints, fields, technologies, components, functions, market needs, actors); coloured nodes and edges are what the model sees. Top row: the full graph (A), no graph (C, only the market need), a random slice of the same size as the M8 slice (R), and the smallest market-need slice (M11). The other panels are the market-need slices (B) for the remaining 12 needs, ordered by size (10 to 51 nodes, 8–41% of the graph).

Table 2: The four representations of the graph given to the model: size of the graph content in the prompt and mean input tokens per call. Ranges for B and R are across the 13 market needs.

| Condition | Nodes | Share of graph | Edges | Input tokens per call (mean, incl. prompt) |
|---|---|---|---|---|
| A | 123 | 100% | 246 | 10,912 |
| B | 10–51 (median 45) | 8–42% | 11–85 | 3,778 |
| R | 10–51, same as B for each market need | 8–42% | 2–52 | 3,143 |
| C | 1 | 1% | 0 | 1,055 |

R matches B in node count and type composition but not in edges. Randomly drawn nodes are less connected to one another than the nodes of a vertical slice, so R contains fewer induced edges than B for every market need. It also uses 17% fewer input tokens. R shares 3–29 nodes with B for the same market need, mostly by chance among the larger slices. A manifest (`slices/manifest.csv`) records, for every file, the node identifiers, counts per type, edge count, token count and SHA-256 hash.

### 3.3 Task and prompt

All calls use one prompt template (`template_prompt.txt`, Appendix B). It asks the model to innovate the product "based on the graph given", encourages radical and wild ideas, and asks it to favour innovations that serve the target market need. Placeholders are filled by literal substitution: product name, target market need identifier and label, operator name and definition, the graph file, and the number of ideas.

- **Operator.** The *Substitute* operator from SCAMPER (Eberle, 1971) in every call: "Identify components and the functions they perform, then replace them with something cheaper or simpler that fulfils the same function equally well or better."
- **Worked example.** One example of an innovation the brief likes, taken from the authors' workshop material (Berengueres, 2014). The door's mesh and the interior lamp both serve the user's need to see the food, so they can be replaced together by a webcam, or by a transparent casing that needs no lamp.
- **Output.** Exactly 5 ideas as JSON. Each idea has a title; the identifiers of existing nodes it was applied to (*anchors*); new nodes (type, label, one-line detail); new edges; and a one-sentence rationale for how it serves the target market need.

The prompt is identical across conditions except for the graph file. In condition C, the "graph given" is only the schema and the target market need.

### 3.4 Generation

- **Model.** `claude-haiku-5-5`, through the Anthropic Messages API, with streaming. No sampling parameters were set; the model does not accept a temperature setting, so all calls use its defaults.
- **Thinking effort.** Set per call through the API's effort parameter, at four of the five available levels: low, medium (the model's default), high and xhigh. Thinking tokens are billed and reported as output tokens.
- **Output limit.** 48,000 tokens (64,000 for xhigh). No call reached its limit.
- **Design.** Five replicates per cell, each a fresh call with the identical prompt:

Table 3 summarises the design.

Table 3: Generation design: conditions run at each thinking effort and on how many market needs, with 5 replicate calls per cell (580 calls in total).

| Conditions | Efforts | Market need node | Calls |
|---|---|---|---|
| A, B | low, medium, high | 13 | 390 |
| A, B | xhigh | 6 (M1, M4, M6, M8, M13, M17) | 60 |
| R, C | high | 13 | 130 |
| **Total** | | | **580** |

The xhigh cells cover 6 market needs, chosen to span small and large slices, to limit cost. R and C were run at high effort only.

- **Parsing.** A response is valid if it parses as JSON with an `ideas` list. 579 of 580 calls were valid with 5 ideas; one returned no parseable ideas and is kept as a failure. In total, the calls produced 2,892 ideas.
- **Logging.** The runner appends one JSON line per call to an append-only log (`runs/rq1_full.jsonl`). Each line holds the run identifier (condition, market need, effort, replicate), the graph file's identifier and hash, the prompt hash, model, output limit, input and output tokens, latency, stop reason, the parsed ideas and the raw response. Runs are resumable, and cells can be added later without rerunning earlier ones.

### 3.5 Judging

Ideas were judged blind by a second, larger model, `claude-sonnet-5-5`, at medium effort, in two ways. In both, the judge saw each idea's **title, new elements (label and detail) and rationale only**. Node identifiers, anchors, edges, condition, effort and run identifiers were removed, so nothing revealed which graph the idea came from. The brief shown to the judge gave the product, target market need, operator, and the instruction that radical, wild ideas serving the market need were wanted.

**Pairwise set judging.** Two idea sets for the same market need and replicate, differing in one factor, were shown side by side. The judge scored each set from 1 to 5 on relevance, novelty, feasibility, specificity and diversity (how different the ideas are from each other), and named the better set or a tie (prompt in Appendix C). Every pair was judged twice, once in each presentation order, and the two verdicts were combined into one result per pair: agreeing verdicts count as a win, opposite verdicts as a tie. If one side had no parsed ideas, the other won without a judge call (one pair per affected comparison). The comparisons judged were:

Table 4 lists the comparisons judged.

Table 4: Pairwise set comparisons sent to the judge. Each pair is two idea sets for the same market need and replicate that differ in one factor.

| Comparison | Effort | Pairs |
|---|---|---|
| A vs B | low, high | 65 each |
| B vs R, B vs C, A vs C | high | 65 each |
| high vs low, within A and within B | — | 65 each |
| xhigh vs high, within A and within B | — | 30 each |
| B at high vs A at low | — | 65 |

**Individual idea scoring.** Each call's five ideas were sent to the judge in one request, with the instruction to rate each idea on its own merits, independently of the others. Each idea was scored from 1 to 5 on relevance, novelty, feasibility and specificity (prompts in Appendix D). An idea's **score** is the mean of the four ratings. All 2,892 ideas were scored; two calls whose first

scoring returned the wrong number of ratings were scored again.

### 3.6 Measures

**Cost.** Input, output and total tokens per call, from the API's usage report.

**Use of the graph** (computed without model calls, from the parsed ideas):

- *anchors outside the market-need slice*: the share of anchors that are existing nodes outside B's slice for that market need;
- *anchors not in the graph*: the share of anchors that match no node identifier in the base graph;
- *ideas echoing the worked example*: ideas whose title or new nodes mention a webcam, camera or transparency;
- *new nodes per idea*, and *words per idea* in the text the judge sees.

**Top tail.** A **top idea** is one whose individual score is at least 4.0. Scores move in steps of 0.25 (one step of a 1–5 rating, averaged over four criteria), so no exact percentile cut exists; 4.0 is the step closest to the top 10% and keeps 12.3% of all ideas (355 of 2,892). As a stricter check, 4.25 keeps the top 2.4% (68 ideas). Target market needs differ widely in how easily they yield top ideas (from 3% to 29% of ideas at 4.0), and the xhigh arms cover only 6 market needs, so every rate is computed per market need and then averaged over the market needs an arm was run on, each market need weighing the same. For each arm:

- **top-idea rate**: the market-need-averaged share of the arm's ideas that are top ideas;
- **calls with a top idea**: the market-need-averaged share of calls producing at least one top idea;
- **tokens per top idea**: the arm's total tokens divided by its number of top ideas;
- **chance of a top idea for a budget** $b$: the probability of at least one top idea when $b$ tokens are spent on whole calls of that arm, averaged over market needs. For need $j$, with $p_j$ the share of calls containing a top idea and $\bar{t}_j$ the mean tokens per call, $k = \lfloor b/\bar{t}_j \rfloor$ calls succeed with probability $1 - (1 - p_j)^k$. Calls are treated as independent. Because 5 calls per market need often give a raw share of exactly 0 or 1, $p_j$ is the Jeffreys estimate $(h_j + \frac{1}{2})/(n_j + 1)$, where $h_j$ of $n_j$ calls contained a top idea.

**Mean idea score.** The market-need-averaged mean individual score per arm, and the mean of each criterion.

### 3.7 Statistics

**Pairwise judging.** The unit is the pair, not the verdict: the two presentation orders of a pair are not independent, so they are combined first. For each comparison:

- the *win rate* of the first-named arm, with ties counting as half a win;
- a Wilson 95% confidence interval;
- an exact two-sided sign test on wins against losses, ties excluded;
- the *order agreement*, the share of pairs that received the same verdict in both orders;
- the mean score difference per criterion.

Whether the market-need slice is no worse than the full graph is tested as non-inferiority with a margin of 45%: B is non-inferior if the lower bound of its win-rate interval against A is above 45%.

To check whether judges favour longer sets, each pairwise record stores the number of words each side showed the judge. The share of decided pairs won by the longer set is reported next to each win rate.

**Top tail.** 95% confidence intervals for the top-idea rate and the call rate come from 2,000 bootstrap resamples. Each resample redraws calls with replacement within each market need, which keeps the five ideas of a call together and the market-need-averaged weighting fixed.

No correction for multiple comparisons was applied. Per-market-need results rest on 5 calls or pairs each and are reported only as patterns.

### 3.8 Plan and pilot

- **Planned before the full run.**
  - The four conditions, including R and C as controls.
  - The A vs B comparison at low and high effort, with the hypotheses that the full graph yields better sets and that B is non-inferior to A at a 45% margin.
  - The effort comparison, high vs low.
  - The hypotheses that B beats R and that a graph beats no graph, to be tested by pairwise judging.
- **Added after the pilot.**
  - Medium and xhigh effort.
  - The B-high vs A-low comparison.
  - Judging in both orders, the length check, and individual idea scoring, which was first used to rank ideas for inspection.
- **Exploratory, defined after seeing the results.**
  - The top-tail analysis: the 4.0 threshold, top-idea rate, tokens per top idea and chance for a budget.
  - The market-need-averaged mean idea scores.

The top-tail measures follow from the purpose of brainstorming (Section 1), but they were chosen after the pairwise results were known and are reported as exploratory. Their conclusions about relevance (B against R) are not confirmed by the planned pairwise test (Section 4).

### 3.9 Pilot

A pilot (2 market needs, M8 and M16; A and B; low, medium and high effort; 3 replicates; 36 calls) tested the pipeline. It led to five changes:

- the output limit was raised from 8,000 to 48,000 tokens, after half the high-effort calls used up their budget thinking and returned no ideas;
- calls were switched to streaming;
- effort was added to the run identifier so cells could be resumed;
- every pair was judged in both orders;
- the length check was added.

The pilot's 17-of-18 advantage for the full graph did not survive at full scale (Section 4).

### 3.10 Human validation

**Expert reading of all ideas.** As a first validation, one of the authors read all 2,892 generated ideas. LLM-ideation studies often assess output linguistically, with embedding-based measures such as the semantic distance between ideas. These are useful for diversity, but they measure how the generated texts relate *to each other*, not to prior art, so the "novelty" or "surprise" they report has little to do with innovation worth in the patent sense. The reading was aimed at exactly that gap, and produced three observations.

1. **The top tier is state of the art, not invention.** Two ideas from the judge's top decile:

   `A_M13_high_r03` — *Solid-State RF Core: Magnetron and HV Stack Replaced by GaN Amplifiers*

   `B_M13_low_r01` — *Magnetron-free solid-state RF cooking head*

   As engineering these are sound; as innovation their value is close to zero. Gallium-nitride power semiconductors made it practical, in the early 2020s, to retire the noisy magnetron and convert the whole RF subsystem to solid state, with gains in weight, noise and vibration, and in the precision of power delivery and control. Dozens of patents apply GaN to microwave ovens and the first commercial products exist; the open problem is cost, at present roughly ten times the magnetron subsystem. A prior-art check would score these ideas at the floor, where the judge's novelty criterion put them near the ceiling.

   The middle of the ranking tells the same story. Five ideas drawn from around the median:

   *Sliding RF-Choke Curtain Replaces the Hinged Door and Push Button*;
   *Photoluminescent cavity glow replaces the 20–25 W lamp*;
   *Sputter-Listening Auto-Stop replaces the timer*;
   *Acoustic Thaw: Ultrasonic Pre-Defrost Replacing Low-Power Cycling*;
   *Phased-Array Field Sweep (No Turntable).*

   Every one of them is prior art or commercially available: sensor-based auto-stop has been standard in consumer ovens for decades, solid-state ovens already steer the field electronically instead of rotating the food, and ultrasound-assisted thawing is established in food processing. From the top of the ranking to its middle, the judge orders the state of the art; it does not separate the new from the known.

2. **Scores are biased towards fit** See a sample from the bottom decile:

   *Ghost Lamp Vision: Replace the Interior Lamp With Microwave-Transparent Walls*;
   *GaN Universal-Mains Power Stage Replaces HV Transformer Stack*;
   *Capacitor-Free 48 V Bus Power Stage*;
   *Direct-Coupled Magnetron Pod (No Waveguide Duct).*

   The GaN power stage here is not materially different from the GaN ideas at the top of the ranking. Part of the gap is defensible, since the judge marks a power-supply substitution down on relevance when it is aimed at a market need it cannot serve; but the same technical move appearing at both ends of the ranking shows that an AI ranker is not entirely consistent, and that its scores reflect the fit between a move and the target market need as much as the move itself.

3. **Diversity.** A topic analysis (Nvivo; not shown) finds the same ideas concentrated in a few recurring topical pathways: solid-state GaN RF generation, transparent or lamp-free enclosures, simplifications of the RF chain, and sensor-based doneness feedback. This should not surprise anyone: each call asked for five ideas under one operator, and further calls revisit the same pathways more often than they open new ones.

Karpathy has described a language model as "a lossy compression of the internet" , and the compressed corpus includes the patent and product literature of this domain. Read this way, the generator behaved as a retrieval system: excellent at surfacing the engineering moves that

bear on a market need, which is also why the ∼11,000-token full-graph prompt and the ∼1,000-token no-graph prompt converge on the same few pathways. The setup cannot be blamed for presenting the state of the art as innovation, if innovation is defined as what no one has patented before. What the experiment measures is therefore the efficiency with which a representation surfaces strong, relevant engineering moves, the raw material of ideation, not patentable novelty; the same reading applies to published systems whose "innovative" output is rated rather than checked against prior art (Section 2.3).

The compression is of what has been written, and what is written about microwave ovens is mostly technology. Reading all the ideas, what is absent is as telling as what is present. Cost appears nowhere as the object of innovation, although the graph carries an explicit low-cost constraint (X12) and cost is the main force selecting microwave innovations: the magnetron has outlived every technically superior alternative because fifty years of manufacturing scale have made it extremely cheap, and that advantage is self-reinforcing, a path dependency that any real substitution must pay its way out of. Nothing in the ideas reflects the IoT turn in consumer electronics; nothing reflects the open-source hardware movement and its community-designed and community-repaired appliances; and nothing uses the fact that most households run a microwave next to a conventional oven, so that several market needs, crisping and browning among them, are already served by the appliance beside it. This knowledge lives in markets, communities and kitchens more than in patents and specification sheets, so the lossy copy retrieves the recorded state of the art and misses the selection environment that decides which of those ideas becomes an innovation. See also section on 'implications for design thinking'.

### 3.11 Reproducibility

The repository (`https://github.com/orioli/IM-LLM`) contains the base graph, the slicer and its manifest, the prompt template, the generation runner, the analysis scripts (`analyze.py`, `top_ideas.py`, `tail_charts.py`, `charts.py`, `human_check.py`), the complete call log with raw responses, all judge outputs, and the derived tables. Every number in Section 4 is regenerated by four commands: `python analyze.py stats`, `python analyze.py summary`, `python top_ideas.py report` and `python tail_charts.py`. Total generation used about 8.6 million tokens.

## 4 Results

### 4.1 Summary

The aim of brainstorming is the rare great idea, not a good average, so the main measure here is the **top tail**: how often an arm produces an idea in the top ∼12% of all ideas, and what each such idea costs.

**RQ1. Which representation yields the most top ideas per token?** (Section 4.3)

- **A relevant slice of the graph finds top ideas fastest per token.** Counted against tokens spent, top ideas accumulate in a straight line for every arm, so each arm has a yield: top ideas per 100k tokens. The market-need slice (B) at medium effort yields 8.8, about three times the full graph (A) at any effort (2.6–3.2). B's top-idea rate is also the highest at equal effort (15% at medium and high, against 10% for no graph and for a random slice of the same size).
- **The efficiency gain holds market need by market need, and is largest for small slices.** The slice is more efficient than the full graph on 10 of 13 market needs (sign test $p = 0.039$), and no graph beats a random slice on 10 of 12. The slice's lead is 3–5× for market needs whose slice has 15–28 nodes, and 1–2× for those with 45–51 nodes (exploratory).

- **No graph is surprisingly efficient.** Its calls are the cheapest, so despite a low top-idea rate its yield (6.4) beats the random slice (4.1) and the full graph (3.1), and is close to the slice at high effort (5.9).

**RQ2. Does the amount or the relevance of the graph matter?** (Section 4.4)

- **The full graph (A) writes the best idea *sets* but does not find more top ideas.** Judged as whole sets, A beats B 64% of the time and no graph 74% of the time. But its top-idea rate is no higher than B's, and its yield is about 3 top ideas per 100k tokens at every effort, half or less of B's: its advantage is the specificity of an average set, not more standout ideas.
- **Relevance vs size is only partly settled.** In the top tail, the market-need slice beats the size-matched random slice (top-idea rate 14.7% vs 9.8%; yield 5.9 vs 4.1 per 100k tokens; the slice's slope is higher in 99% of paired bootstrap draws). Judged as whole sets, the two cannot be told apart (B over R 54%, $p = 0.55$). What a relevant slice adds shows up in the best ideas, not in the average set.
- **A graph helps, clearly so when it is the full graph.** A beats C on whole sets (74%, $p < 0.001$); B beats C less clearly (62%, $p = 0.053$) but has a higher top-idea rate (14.7% vs 10.2%).

**RQ3. How does spending tokens on context compare with spending them on thinking effort?** (Section 4.5)

- **Thinking effort is the strongest lever on quality, but not on yield.** High beats low in about 93% of set comparisons, and xhigh produces the highest top-idea rates of all (19–21%). But each call costs proportionally more, so yield does not rise with effort: for the full graph it stays at about 3 per 100k tokens from low to xhigh, and for the slice it peaks at medium (8.8) and falls at high (5.9) and xhigh (3.1).

**Exploratory: persona** (Section 4.6)

- **Persona prompting did not raise efficiency (exploratory follow-up).** On three market needs, neither a lateral-thinker nor a patent-focused persona beat the plain innovation engineer per token (6.3 top ideas per 100k tokens; patent persona 5.5, lateral thinker 4.3, lower than the engineer's in $> 99.9\%$ of bootstrap draws); the lateral thinker was the least efficient, especially without a graph. The market-need slice helped every persona; whether it helps some personas more than others could not be resolved.

## 4.2 Setup, cost and validity

### 4.2.1 Overview

Table 5 summarises the experimental setup.

Table 5: Experimental setup at a glance.

| Item | Value |
|---|---|
| Product graph | `microwave_base_graph.yaml`: 123 nodes, 246 edges |
| Target needs | 13: M1, M3, M4, M5, M6, M7, M8, M11, M12, M13, M14, M16, M17 (M2, M9, M10, M15 excluded: no functions address them, so their slice is empty) |

Table 5 (continued)

| Item | Value |
|---|---|
| Conditions | **A** = full graph · **B** = vertical slice of the target market need (functions, components, technologies, fields, linked constraints and actors) · **R** = random slice with the same number of nodes per layer as B · **C** = no graph (product description and target market need only) |
| Thinking effort | A and B: low, medium, high on all 13 market needs, xhigh on 6 market needs (M1, M4, M6, M8, M13, M17) · R and C: high on all 13 market needs |
| Replicates | 5 per cell |
| Generation calls | 580: A and B 450 (13 market needs × 2 conditions × 3 efforts × 5 reps, plus 6 market needs × 2 × 5 at xhigh); R and C 130 (13 × 2 × 5) |
| Prompt | `template_prompt.txt`: 5 radical ideas for the target market need, operator SUBSTITUTE, worked example (door holes + lamp → webcam / transparent casing), JSON output |
| `max_tokens` | 48,000 (64,000 for xhigh); no call was truncated |
| Sampling | model defaults (Haiku 5.5 accepts no temperature setting) |
| Pairwise judging | A vs B at low and high effort; B vs R, B vs C, A vs C at high effort; effort steps within A and B; B-high vs A-low. Blind comparison of two idea sets for the same market need and replicate; no condition labels or node IDs shown; each pair judged in both presentation orders and the two verdicts combined (agreeing verdicts = a win, split = a tie); scores 1–5 on relevance, novelty, feasibility, specificity, diversity |
| Individual scoring | every idea scored on its own, blind, 1–5 on relevance, novelty, feasibility, specificity; idea score = mean of the four (all 2,892 ideas) |
| Statistics | pairwise: per pair (65 pairs per comparison, 30 for xhigh), ties count half, Wilson 95% CI, exact sign test · tail: rates averaged over market needs (each market need weighs the same), 95% CI by resampling calls within each market need (2,000 resamples) |

### 4.2.2 Cost and validity

Figure 4 shows tokens per call by condition and effort; Figure 5 sets the four representations side by side at high effort, input tokens against mean idea score.

Table 6 gives the means per call.

Table 6: Cost and validity by representation and effort: number of calls, share of responses parsed, mean tokens per call (input, output, total) and mean words per idea.

| Condition | Effort | Calls | Parse rate | Input tokens | Output tokens | Total tokens | Words per idea |
|---|---|---|---|---|---|---|---|
| A (Full) | low | 65 | 100% | 10,912 | 3,331 | 14,243 | 91 |
| A (Full) | medium | 65 | 100% | 10,912 | 5,295 | 16,208 | 99 |
| A (Full) | high | 65 | 100% | 10,912 | 11,163 | 22,076 | 108 |
| A (Full) | xhigh | 30 | 100% | 10,912 | 28,346 | 39,258 | 117 |
| B (Slice) | low | 65 | 100% | 3,778 | 2,849 | 6,627 | 89 |
| B (Slice) | medium | 65 | 100% | 3,778 | 4,772 | 8,550 | 98 |
| B (Slice) | high | 65 | 98.5% | 3,778 | 8,628 | 12,407 | 100 |
| B (Slice) | xhigh | 30 | 100% | 4,135 | 25,885 | 30,020 | 117 |
| R (Rnd) | high | 65 | 100% | 3,143 | 8,778 | 11,921 | 98 |
| C (No graph) | high | 65 | 100% | 1,055 | 6,939 | 7,994 | 106 |

xhigh was run on 6 of the 13 market needs; those market needs have larger slices on average, which is why B's xhigh input is higher.

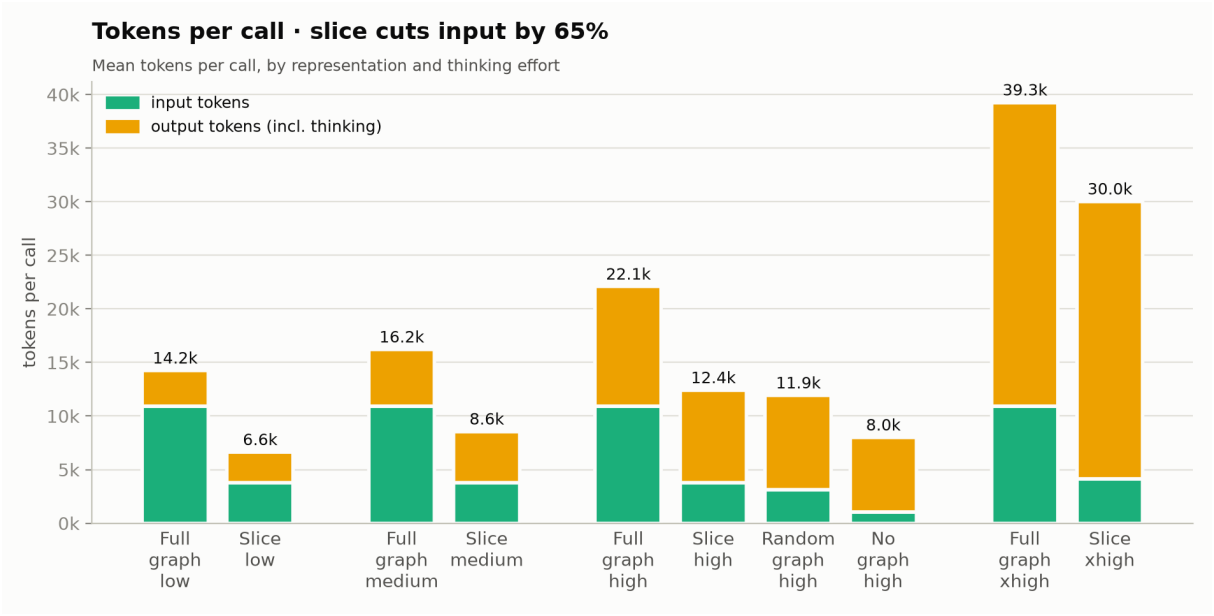



Figure 4: Mean tokens per call by representation and effort, split into input (the prompt, most of it the graph file) and output (thinking and answer). The full graph pays for its context on every call; thinking effort drives the output for every representation.

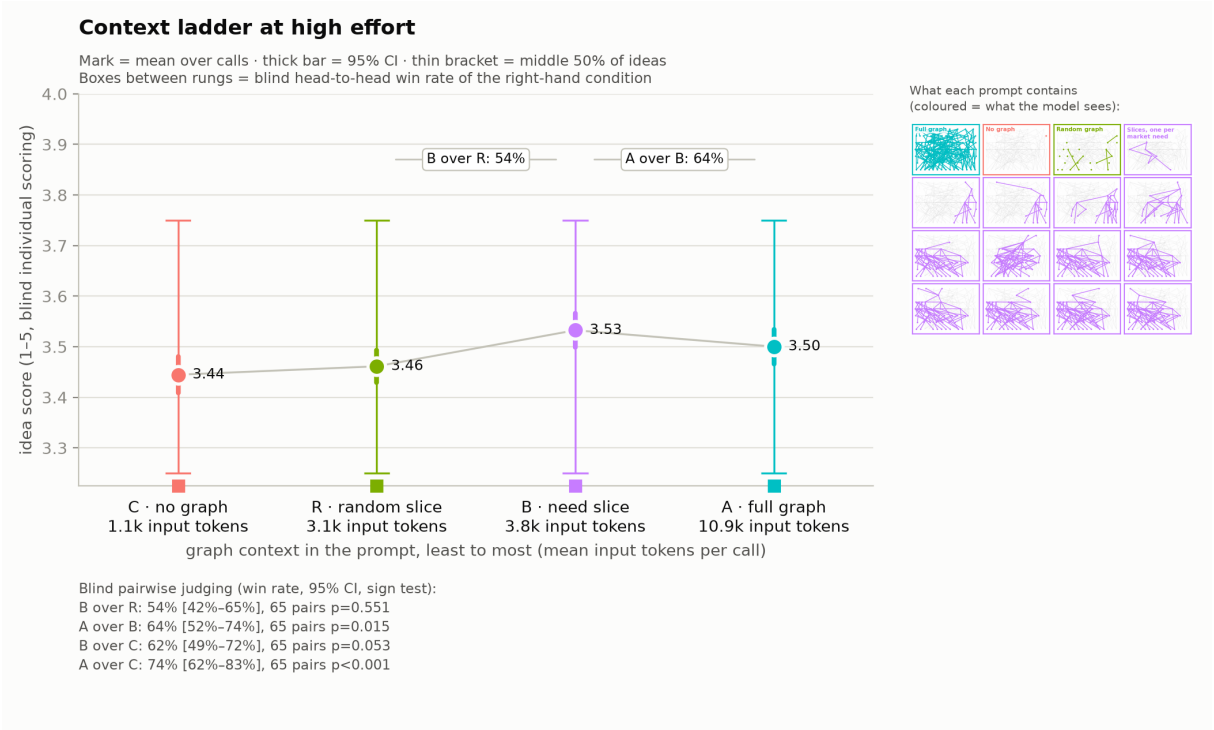



Figure 5: The context ladder at high effort: no graph (C), random slice (R), market-need slice (B) and full graph (A), in order of input size, against mean idea score, with the head-to-head set win rates between rungs. Note that innovation does not reward 'average' but outliers.

- **Input:** the slice cuts input tokens by 65% on average (slices range from 10 to 51 nodes, 8–42% of the graph). R matches B's node counts per layer but uses 17% fewer input tokens, because random nodes have shorter labels and fewer linking edges; the size match is close but not exact.
- **Total:** the slice saves 53% of total tokens at low effort, 47% at medium, 44% at high and 24% at xhigh. Output grows steeply with effort and soon dominates the bill, so the input saving matters less and less.
- **Output grows with context:** at high effort, output is 6.9k tokens with no graph, 8.6–8.8k with a slice (B or R) and 11.2k with the full graph. More context makes the model think and write more, a cost beyond the prompt itself.
- **Output by effort:** each step up costs more than the last: +1,900–2,000 tokens from low to medium, +3,900–5,900 from medium to high, about +17,000 from high to xhigh.
- **Validity:** 579 of 580 calls returned valid JSON with 5 ideas; none hit the token limit.

#### 4.2.3 How the ideas use the graph

Table 7 shows where the ideas anchor in the graph and how much they add to it.

Table 7: How the ideas use the graph, by representation and effort: where ideas anchor, how often they repeat the prompt's worked example, and how many new nodes they propose.

| Condition | Effort | Anchors outside the market need's own slice | Anchors to nodes not in the graph | Ideas repeating the worked example (of 5) | New nodes per idea |
|---|---|---|---|---|---|
| A (Full graph) | low | 53% | 0% | 0.95 | 2.8 |
| A (Full graph) | medium | 49% | 0% | 0.69 | 3.1 |
| A (Full graph) | high | 48% | 0% | 0.54 | 3.4 |
| A (Full graph) | xhigh | 40% | 0% | 0.33 | 3.6 |
| B (Slice) | low | 0% | 0% | 0.86 | 2.9 |
| B (Slice) | medium | 0% | 0% | 0.48 | 3.1 |
| B (Slice) | high | 0% | 0% | 0.55 | 3.3 |
| B (Slice) | xhigh | 0% | 0% | 0.40 | 3.9 |
| R (Random graph) | high | 42% | 0% | 0.51 | 3.3 |
| C (No graph) | high | — | 48% | 0.37 | 4.4 |

- With the full graph, about half of all anchors lie outside the target market need's slice: A uses its extra context to reach across the product.
- With a random slice, 42% of anchors fall outside the market need's slice, because most of what R shows is unrelated to the market need.
- With no graph, the model invents node IDs for half of its anchors and adds more new nodes per idea (4.4): it builds its own picture of the product instead of working on the given one.
- Repetition of the worked example falls with effort, and is lowest with no graph.

### 4.3 RQ1: Which representation yields the most top ideas per token?

**Answer.** The market-need slice. At medium effort it yields 8.8 top ideas per 100k tokens, about three times the full graph at any effort (2.6–3.2), and it is more efficient than the full graph on 10 of 13 market needs. At high effort, where all four representations were run, yields rank no graph (6.4) $\approx$ market-need slice (5.9) $>$ random slice (4.1) $>$ full graph (3.1).

A **top idea** scores $\geq 4.0$ on the individual blind scoring. Scores move in 0.25 steps, so an exact top 10% does not exist: $\geq 4.0$ keeps the top 12.3% of all ideas (355 of 2,892), $\geq 4.25$ the

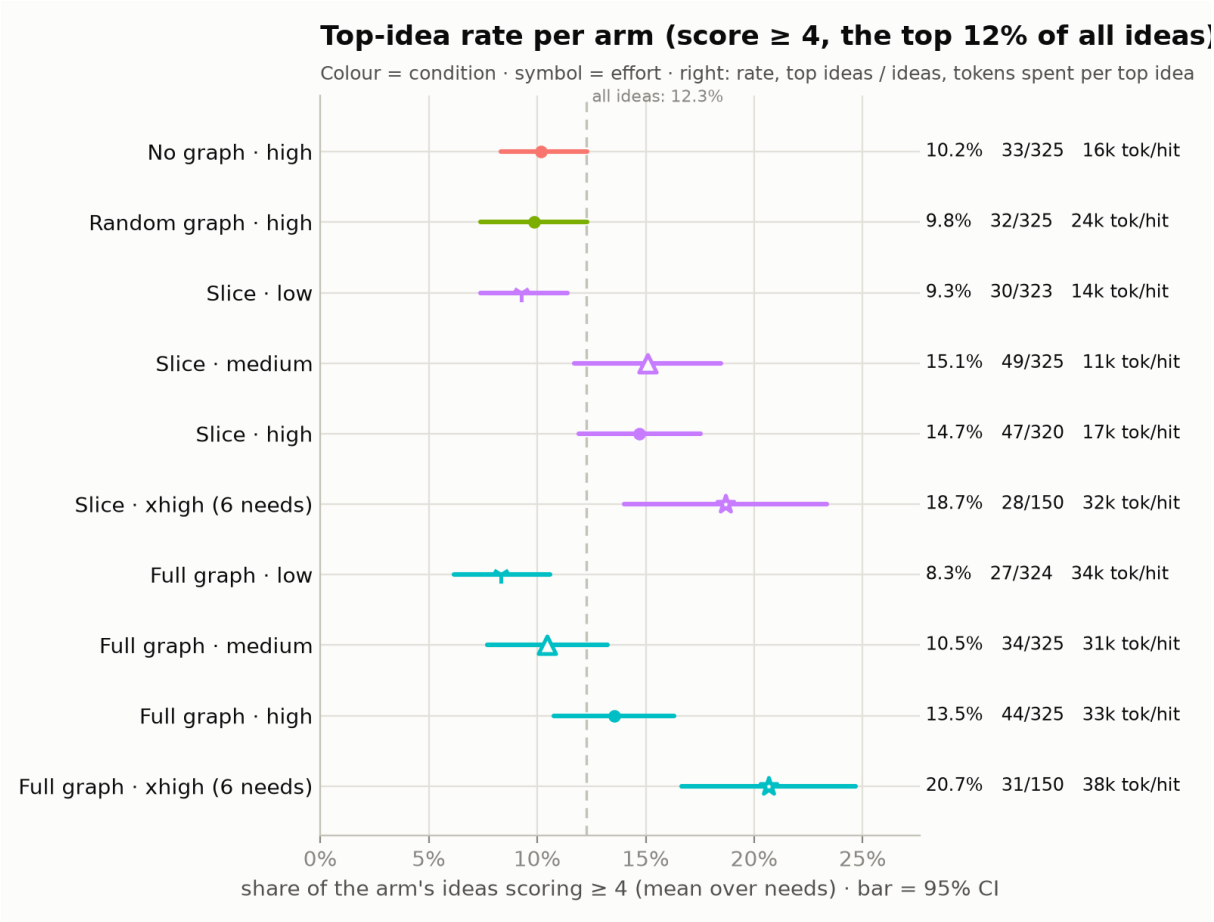



Figure 6: Share of each arm's ideas that are top ideas (score $\geq 4.0$, the top 12% of all ideas), averaged over market needs, with 95% intervals from resampling calls within market need. This is quality only; cost per call is not taken into account here.

top 2.4% (68). Market needs differ a lot in how easy a top idea is (29% of M14's ideas reach 4.0, 3% of M11's), so every rate is the mean of per-market-need rates.

### 4.3.1 Top-idea rate (score $\geq 4.0$)

Figure 6 shows the rates with their intervals; Table 8 gives them with their cost.

Table 8: Top-idea rate per arm (share of ideas scoring $\geq 4.0$), averaged over market needs with 95% bootstrap intervals, with cost per call and per top idea. Bold: lowest cost per top idea.

| Arm | Needs | Top ideas / ideas | Top-idea rate | 95% CI | Calls with $\geq 1$ top idea | Tokens per call | Tokens per top idea |
|---|---|---|---|---|---|---|---|
| C (No graph) · high | 13 | 33 / 325 | 10.2% | 8.3–12.3% | 48% | 7,994 | 15,746 |
| R (Random graph) · high | 13 | 32 / 325 | 9.8% | 7.7–12.3% | 45% | 11,921 | 24,215 |
| B (Slice) · low | 13 | 30 / 323 | 9.3% | 7.4–11.4% | 37% | 6,627 | 14,359 |
| B (Slice) · medium | 13 | 49 / 325 | 15.1% | 11.7–18.5% | 52% | 8,550 | **11,342** |
| B (Slice) · high | 13 | 47 / 320 | 14.7% | 11.9–17.5% | 56% | 12,384 | 16,838 |

Table 8 (continued)

| Arm | Needs | Top ideas / ideas | Top-idea rate | 95% CI | Calls with ≥ 1 top idea | Tokens per call | Tokens per top idea |
|---|---|---|---|---|---|---|---|
| B (Slice) · xhigh | 6 | 28 / 150 | 18.7% | 14.0–23.3% | 73% | 30,020 | 32,164 |
| A (Full graph) · low | 13 | 27 / 324 | 8.3% | 6.2–10.5% | 37% | 14,243 | 34,289 |
| A (Full graph) · medium | 13 | 34 / 325 | 10.5% | 8.0–13.2% | 45% | 16,208 | 30,985 |
| A (Full graph) · high | 13 | 44 / 325 | 13.5% | 10.5–16.3% | 57% | 22,076 | 32,612 |
| A (Full graph) · xhigh | 6 | 31 / 150 | 20.7% | 16.7–24.7% | 80% | 39,258 | 37,991 |

- **Full graph vs slice:** at every effort, B's top-idea rate is equal to or higher than A's, at about half the tokens per top idea.
- **Cheapest top ideas:** B at medium effort (11k tokens each), then B at low (14k), C (16k) and B at high (17k). A costs 31–38k at every effort.

#### 4.3.2 Yield: top ideas per 100k tokens

The top-idea rate says how good an arm's ideas are; the yield says how fast it finds top ideas for the money. Counting top ideas against the tokens spent on an arm's calls (taken in replicate order, cycling through the market needs) gives a running count that grows in a straight line for every arm (Figure 7). The **yield** is the slope of that line: all of the arm's top ideas divided by all its tokens, per 100k tokens, with a 95% CI from 2,000 resamples of calls within each market need. Unlike a probability, a count does not level off, so arms are compared directly by slope. Table 9 gives the yields.

Table 9: Yield per arm, the main measure: top ideas per 100k tokens with 95% bootstrap intervals, ranked from most to least efficient.

| Arm | Top ideas | Tokens spent | Yield (top ideas per 100k tokens) | 95% CI |
|---|---|---|---|---|
| B (Slice) · medium | 49 | 0.56M | **8.8** | 7.0–10.7 |
| B (Slice) · low | 30 | 0.43M | 7.0 | 5.4–8.4 |
| C (No graph) · high | 33 | 0.52M | 6.4 | 5.1–7.7 |
| B (Slice) · high | 47 | 0.79M | 5.9 | 4.9–7.0 |
| R (Random graph) · high | 32 | 0.77M | 4.1 | 3.1–5.2 |
| A (Full graph) · medium | 34 | 1.05M | 3.2 | 2.5–4.1 |
| B (Slice) · xhigh (6 market needs) | 28 | 0.90M | 3.1 | 2.4–3.9 |
| A (Full graph) · high | 44 | 1.43M | 3.1 | 2.4–3.7 |
| A (Full graph) · low | 27 | 0.93M | 2.9 | 2.2–3.7 |
| A (Full graph) · xhigh (6 market needs) | 31 | 1.18M | 2.6 | 2.1–3.1 |

- **The market-need slice is the most efficient representation at every effort up to high.** B at medium yields 8.8 top ideas per 100k tokens, 2.8 times A at medium (3.2); at high, 5.9 against 3.1. The intervals do not overlap.
- **How far apart the slopes are** (Figure 8, chance that each arm's slope beats the next lower arm's): B · medium over B · low 0.93; B · low over C 0.74; C over B · high 0.69; B · high over R 0.99; R over A · medium 0.91; among the full-graph arms and B · xhigh

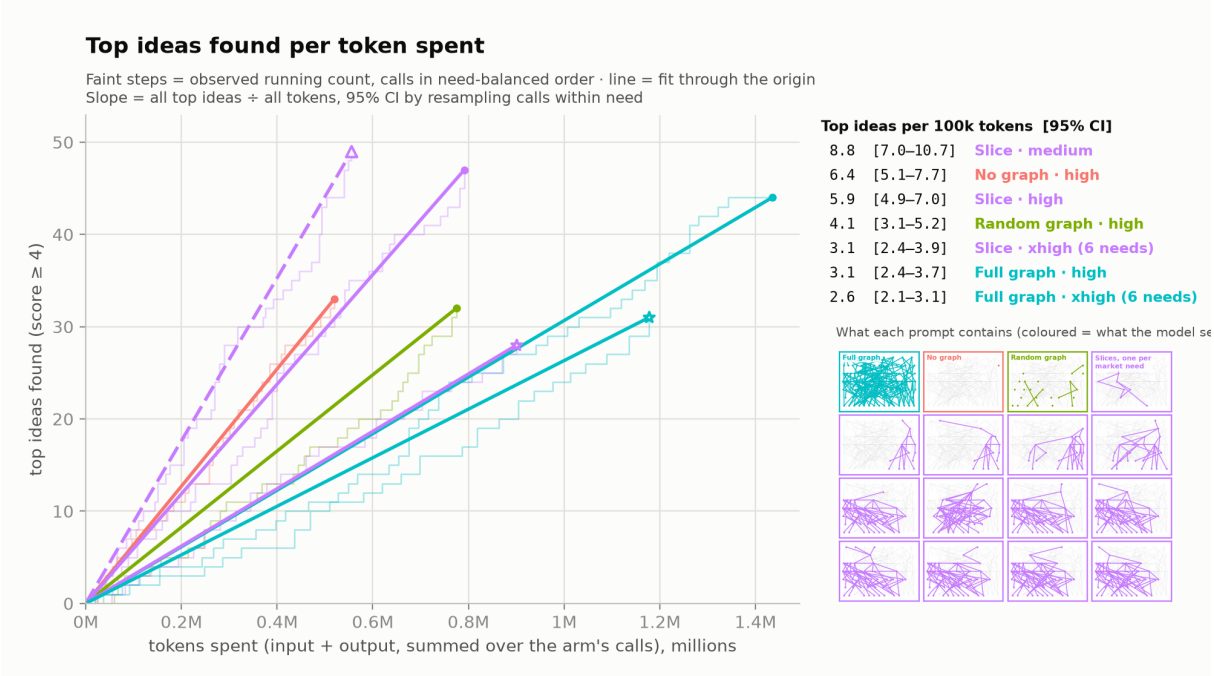



Figure 7: Top ideas (score $\geq$ 4.0) found against tokens spent, per arm, with calls taken in replicate order. Each arm accumulates top ideas at a near-constant rate, so its slope is its yield (top ideas per 100k tokens): the steeper the line, the more efficient the arm. The market-need slice at medium effort climbs fastest; the full-graph arms are the flattest at every effort.

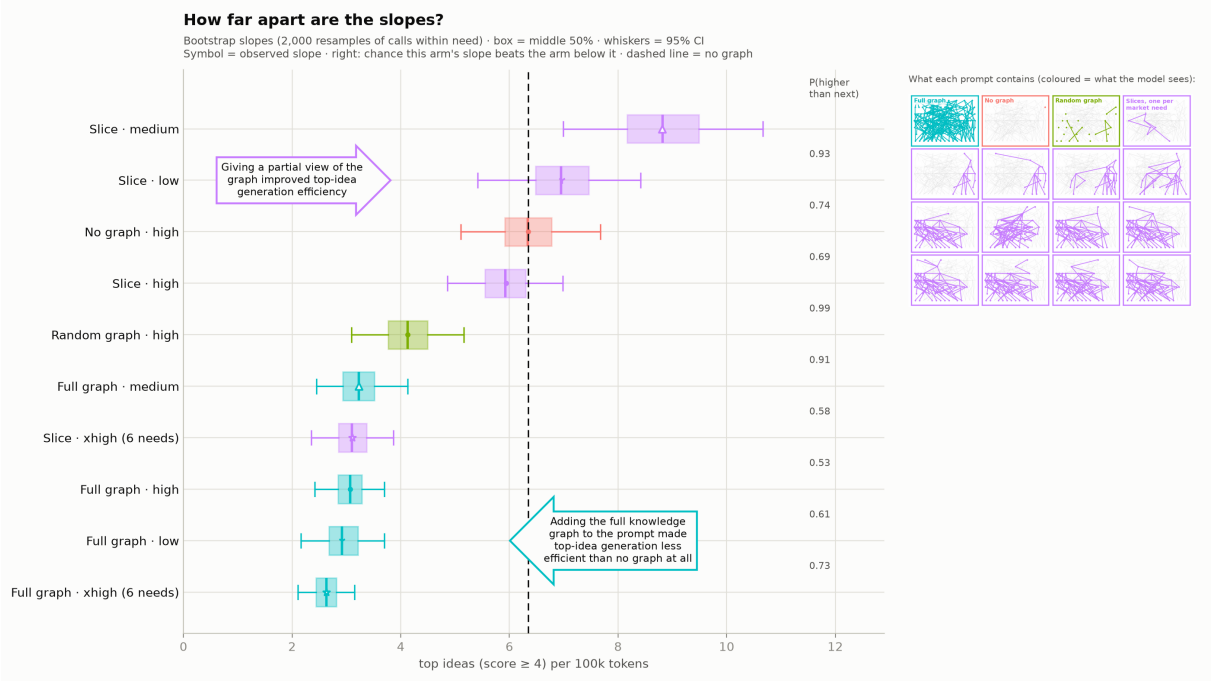



Figure 8: How certain the yields are: for each arm, a box plot of 2,000 bootstrap slopes (calls resampled within market need; box = middle 50%, whiskers = 95% interval). The dashed line marks the control, no knowledge graph in the prompt (C, high effort). Arms whose boxes lie to the left of the line find top ideas less efficiently than giving the model no graph at all; this includes every full-graph arm, the random slice of the graph, and the slice of graph dangling from a given market-need slice. When using a graph slices, only setting at low and medium effort underperfrom the control ”C”

0.53–0.73. Four tiers: B at medium (8.8); B at low and high with C (5.9–7.0); R (4.1); all full-graph arms and B · xhigh (2.6–3.2).

- **No graph:** C's low top-idea rate (10.2%) is offset by its cheap calls (8.0k tokens), giving a yield of 6.4, above R and A and within the interval of B at high effort. Its weakness is coverage: with no graph, M1 produced no top idea in 5 calls (A and B: 2–3).
- **At the stricter threshold ($\geq 4.25$)** the order is similar but the counts are small (2–11 per arm): B · medium 1.8 per 100k tokens (1.1–2.7), A · medium 1.0, B · low 0.9, B · high 0.9, A · xhigh 0.85, B · xhigh 0.8, C 0.8, A · high 0.7, R 0.4, A · low 0.2.
- **On the 6 xhigh market needs only:** B · medium 9.0, B · low 7.4, B · high 6.4, C 6.1, R 4.2, A · low 3.3, B · xhigh 3.1, A · medium 2.9, A · high 2.8, A · xhigh 2.6. The ranking is the same as on all market needs.

### 4.3.3 Yield per market need

To check that the yield ranking is not driven by a few easy market needs, yields were computed for each market need separately (5 calls per arm and market need), and arms were compared by the number of needs on which each is more efficient (exact two-sided sign test, ties excluded; `yield_per_market_need.xlsx`).

Table 10 gives the yield per market need at high effort (top ideas per 100k tokens; B's slice size in nodes), and Table 11 the sign tests.

Table 10: Yield per market need at high effort (top ideas per 100k tokens), ordered by the size of the market-need slice. The last column is the slice's yield divided by the full graph's.

| Need | Slice nodes | B market-need slice | C no graph | R random slice | A full graph | $B \div A$ |
|---|---|---|---|---|---|---|
| M11 | 10 | 2.1 | 0.0 | 2.1 | 1.7 | 1.2× |
| M14 | 15 | 17.9 | 15.5 | 10.6 | 8.2 | 2.2× |
| M16 | 15 | 9.2 | 5.1 | 2.5 | 3.0 | 3.1× |
| M8 | 25 | 13.4 | 10.1 | 7.6 | 2.9 | 4.7× |
| M6 | 28 | 11.8 | 7.4 | 5.8 | 3.4 | 3.5× |
| M7 | 45 | 5.5 | 12.8 | 5.8 | 2.8 | 2.0× |
| M13 | 45 | 6.0 | 5.0 | 3.1 | 2.7 | 2.2× |
| M3 | 47 | 4.1 | 10.0 | 4.8 | 4.5 | 0.9× |
| M12 | 47 | 3.1 | 2.6 | 1.6 | 2.8 | 1.1× |
| M1 | 48 | 2.6 | 0.0 | 0.0 | 3.2 | 0.8× |
| M4 | 51 | 4.3 | 6.8 | 4.1 | 1.9 | 2.3× |
| M5 | 51 | 0.0 | 0.0 | 1.5 | 0.0 | — |
| M17 | 51 | 2.9 | 7.9 | 6.1 | 2.8 | 1.0× |

Table 11: Arm-versus-arm comparisons across market needs: number of needs on which each arm has the higher yield, with exact two-sided sign tests (ties excluded). Bold: $p < 0.05$.

| Comparison | First arm more efficient | Second more efficient | Equal | Sign test p |
|---|---|---|---|---|
| B vs A, high effort | 10 market needs | 2 | 1 | **0.039** |
| B vs A, medium effort | 10 | 3 | 0 | 0.092 |
| C vs R (no graph vs random slice) | 10 | 2 | 1 | **0.039** |
| R vs A | 10 | 3 | 0 | 0.092 |
| C vs A | 9 | 3 | 1 | 0.146 |
| B vs R | 9 | 4 | 0 | 0.267 |
| B vs C | 8 | 4 | 1 | 0.388 |

- **The slice's efficiency advantage over the full graph holds market need by market**

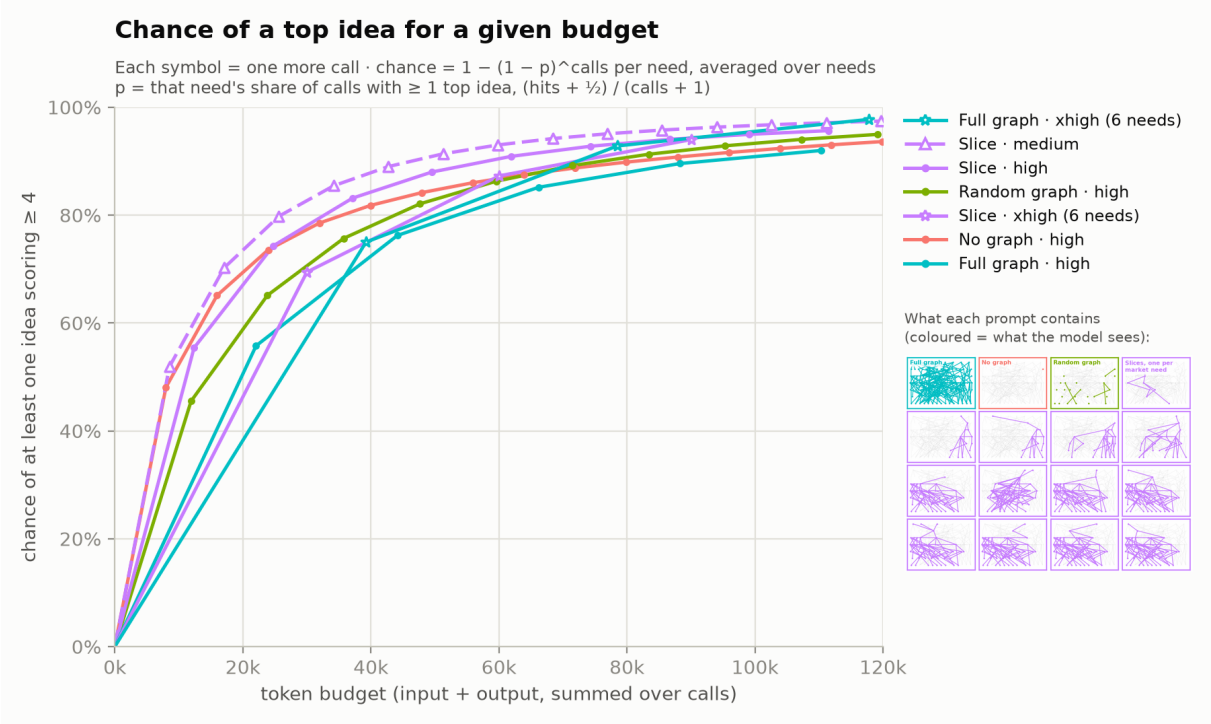



Figure 9: Chance of at least one top idea for a given token budget, per arm, from whole calls and a Jeffreys estimate of each market need's success rate. Every curve rises towards 1 as the budget grows, so this view compresses differences at large budgets; yield (Figure 7) is the main measure.

**need:** B is more efficient on 10 of 13 market needs at high effort (p = 0.039) and at medium effort (p = 0.09). The full graph is ahead only on M1 and M3, by small margins.

- **Single needs are noisy:** each value rests on 5 calls and 0–10 top ideas.

#### 4.3.4 Chance of at least one top idea for a given budget (secondary)

Figure 9 shows the curves.

This measure answers a practical question, how likely one budget is to produce at least one top idea, but every arm's chance approaches 100% as the budget grows, so it compresses differences at large budgets and can suggest that spending alone guarantees a top idea. The yield above is the better basis for comparing arms. For a token budget spent on whole calls of one arm, the chance that at least one idea scores $\geq 4.0$, averaged over market needs. Within a market need, calls are treated as independent: $k$ calls succeed with $1-(1-p)^k$, where $p$ is the market need's share of calls with a top idea, estimated as $(\text{hits}+\frac{1}{2})/(\text{calls}+1)$ because 5 calls per market need often give exactly 0 or 1.

Table 12 gives the chance at three budgets.

Table 12: Chance of at least one top idea for a given token budget, per arm. In brackets: the number of whole calls that budget buys.

| Arm | 25k tokens | 50k tokens | 100k tokens |
|---|---|---|---|
| No graph · high | 74% (3 calls) | 84% (6) | 92% (12) |
| Random graph · high | 65% (2) | 82% (4) | 93% (8) |
| Slice · low | 64% (3) | 81% (7) | 91% (15) |
| Slice · medium | 70% (2) | **89%** (5) | **96%** (11) |
| Slice · high | 74% (2) | 88% (4) | 95% (8) |
| Slice · xhigh (6 market needs) | — (0) | 69% (1) | 94% (3) |

Table 12 (continued)

| Arm | 25k tokens | 50k tokens | 100k tokens |
|---|---|---|---|
| Full graph · low | 39% (1) | 67% (3) | 86% (7) |
| Full graph · medium | 46% (1) | 79% (3) | 94% (6) |
| Full graph · high | 56% (1) | 76% (2) | 90% (4) |
| Full graph · xhigh (6 market needs) | — (0) | 75% (1) | 93% (2) |

- **B at medium and high effort lead at 50k and 100k tokens,** and B-medium is at or near the top at every budget.
- **No graph is a good choice for small budgets** because its calls are cheapest, but it rises more slowly at large budgets because of the market needs where it finds nothing (M1); M5 and M11 are hard for every arm.
- **The full graph trails at every budget:** its calls are expensive and its top-idea rate is no higher than B's.

#### 4.3.5 Stricter threshold and matched market needs

- **Score** $\geq 4.25$ **(top 2.4%):** same direction, too few hits per arm (2–11) to separate arms. A · xhigh 6.7% and B · xhigh 4.7% (6 market needs), A · medium 3.4%, B · medium 3.1%, A · high 3.1%, B · high 2.2%, C 1.2%, B · low 1.2%, R 1.0%, A · low 0.6%. Tokens per top idea are lowest for B · medium (56k) and highest for R (about 260k) and A · low (463k); yields are listed in the yield section above.
- **Only the 6 xhigh market needs** (`--match-needs`): B · medium and B · high 16.7%, B · xhigh 18.7%, A · high 12.7%, A · xhigh 20.7%, R 10.7%, C 10.0%. On equal market needs, B gains little from xhigh over medium or high, while A gains a lot; A needs the extra thinking to make use of its larger context.

#### 4.3.6 The top ideas

The 10 highest-scoring of all 2,892 ideas (`results/top_ideas.md`), all scoring 4.50: 5 from A (high ×2, medium ×2, xhigh), 4 from B (high ×2, low, xhigh) and 1 from R (high). The bottom 10 are dominated by B at low effort (6 of 10).

- **The best ideas converge on two themes from many arms:** replacing the magnetron and high-voltage chain with a solid-state GaN RF source (for M13 energy efficiency, and from R for M7 safety), and solid-state 27 MHz or 915 MHz RF thawing for M4. Arms differ in how often they reach these ideas, not in whether they can.
- **The same technical theme scores low when it does not serve the target market need** (e.g. a GaN power stage proposed for M12, healthy diet): the judge rewards fit to the market need, not technical appeal alone.
- Many ideas tie at the top, so the ranked list identifies top-scoring ideas rather than a strict order.

### 4.4 RQ2: Does the amount or the relevance of the graph matter?

**Answer.** Relevance, not amount. More graph buys better average sets: the full graph wins 64–65% of set comparisons against the market-need slice, mainly on specificity, but its top-idea rate is no higher, so at about twice the tokens per call it finds fewer top ideas per token. Relevant structure helps in the top tail: the market-need slice out-yields a random slice of the same size in 99% of paired bootstrap draws, although whole-set judging cannot tell the two apart. Irrelevant structure is worse than none: no graph is more efficient than the random slice on 10 of 12 market needs.

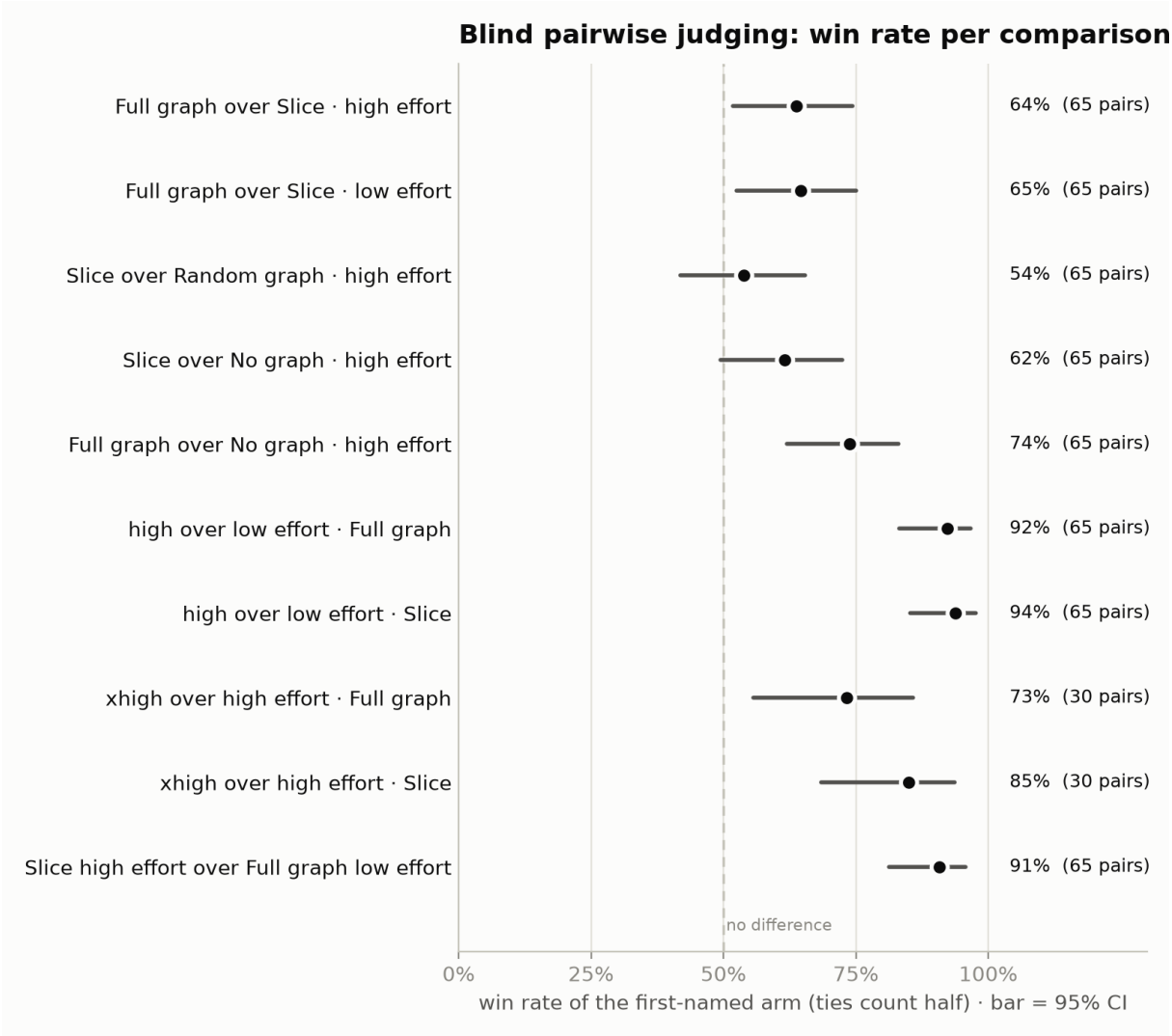



Figure 10: Every pairwise set comparison: the first arm's win rate with a 95% Wilson interval (each pair judged in both presentation orders, ties excluded). Values above 50% favour the first arm.

### 4.4.1 Amount: full graph vs market-need slice

Figure 10 shows every pairwise set comparison in this section and in Section 4.5; Figure 11 shows how the full graph and the slice differ on each criterion. Table 13 gives the win rates and Table 14 the differences per criterion.

Table 13: Pairwise set judging, full graph versus market-need slice at equal effort: wins, win rate with 95% Wilson interval, sign test, and agreement between the two presentation orders.

| Effort | Pairs | Full graph wins | Slice wins | Ties | Full graph win rate | 95% CI | Sign test p | Order agreement |
|---|---|---|---|---|---|---|---|---|
| low | 65 | 37 | 18 | 10 | 65% | 53–75% | 0.015 | 85% |
| high | 65 | 34 | 16 | 15 | 64% | 52–74% | 0.015 | 77% |

Mean score difference, Full graph minus Slice (1–5 scale):

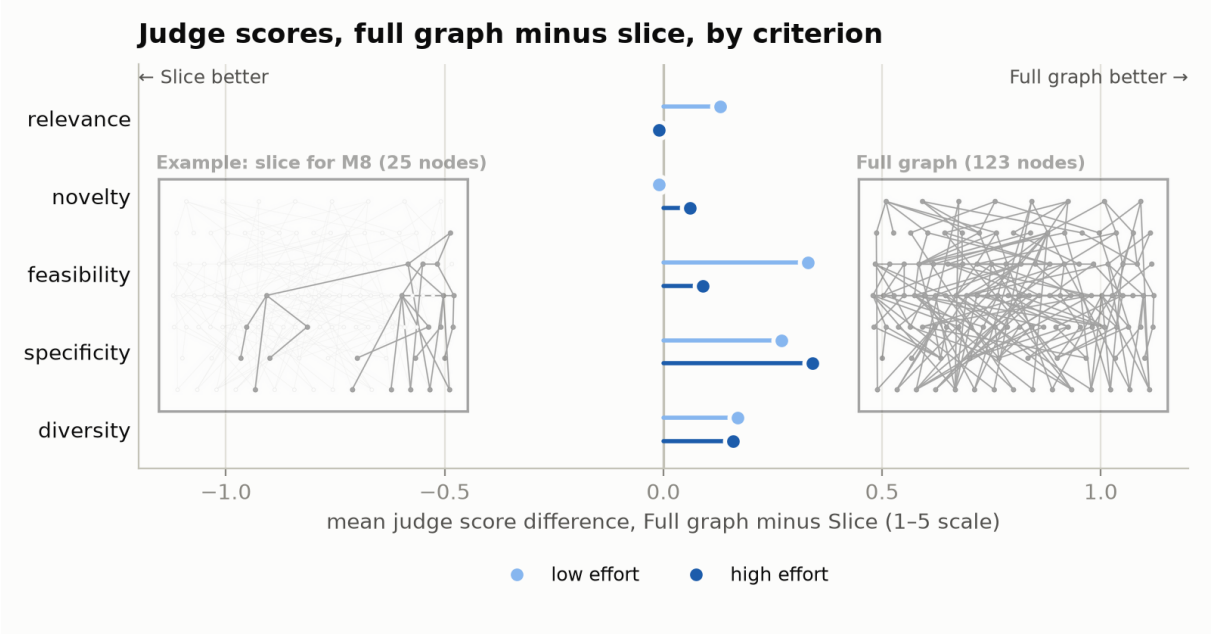



Figure 11: Full graph minus market-need slice (A − B) on each set-judging criterion, by effort. The full graph's advantage is mainly specificity.

Table 14: Full graph minus market-need slice: mean difference in set scores per criterion (1–5 scale), by effort.

| Effort | Relevance | Novelty | Feasibility | Specificity | Diversity |
|---|---|---|---|---|---|
| low | +0.13 | −0.01 | +0.33 | +0.27 | +0.17 |
| high | −0.01 | +0.06 | +0.09 | +0.34 | +0.16 |

- **The full graph yields better sets:** A wins 65% of set comparisons at low effort and 64% at high (p = 0.015 at both).
- The full graph's advantage on whole sets is real but modest, mainly in specificity; novelty and relevance are about equal.
- **The slice is not shown to be non-inferior on sets:** B's win rate is 35–36%, with the lower confidence bound (25–26%) well below the 45% non-inferiority margin. On the tail measure, however, B is not worse than A at any effort.
- **The effect varies by market need:** A swept 5–0 for M17 and M6 at high effort and M11 at low; B led for M1 at low effort, M3 at both levels and M8 at high. With 5 pairs per market need, no single market need is significant.

**Slice size (exploratory).** The less of the graph the slice contains, the larger its advantage. B's lead over A is largest where the slice is small: 3–5× for M6, M8 and M16 (15–28 nodes), against 0.8–2.3× for the market needs with 45–51 nodes (Figure 12). Across market needs, the log of B's yield ratio over A correlates with slice size at $r = -0.46$ at high effort ($n = 12$, $p \approx 0.13$) and $r = -0.61$ at medium effort ($n = 12$, $p \approx 0.04$). M11, the smallest slice (10 nodes), is an exception at 1.2×; it is also one of the two hardest market needs for every arm. A smaller slice saves more tokens and is more focused; with 13 market needs and 5 calls each this is a pattern to test, not an established result.

### 4.4.2 Relevance in the top tail: market-need slice, random slice and no graph

The market-need slice (B) and the random slice (R) have the same number of nodes per layer, so comparing them separates relevance from amount; no graph (C) is the floor.

- **Top-idea rate, B vs R:** at the same effort and nearly the same size, the market-need slice yields top ideas at 14.7% against 9.8% for the random slice; the confidence intervals just touch. The whole-set judging does not confirm this (Section 4.4.3).
- **Yield, B vs R:** at high effort, B yields 5.9 and R 4.1; the 95% intervals just touch (4.9–7.0 against 3.1–5.2), but pairing the two arms' bootstrap draws, B's slope is the higher in 99%

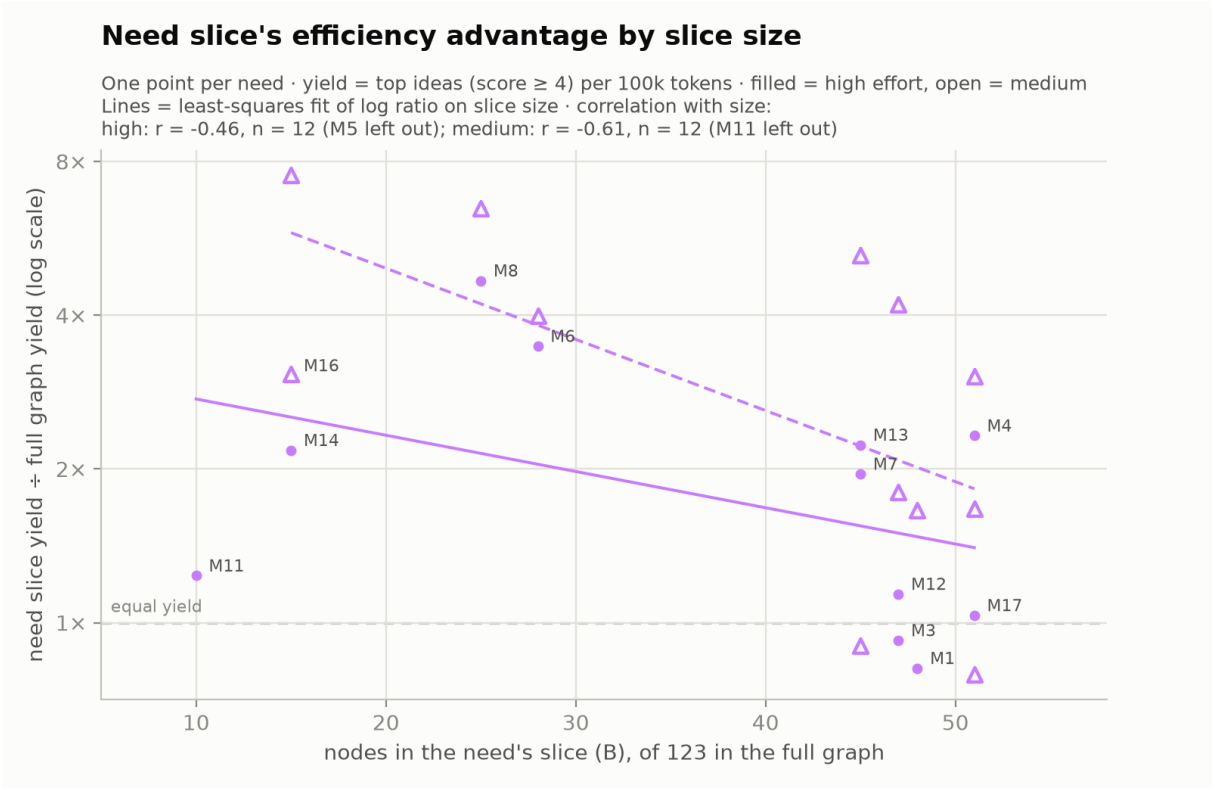



Figure 12: Per market need: the market-need slice's yield divided by the full graph's, against the number of nodes in the slice (log scale; market needs with zero yield omitted).

Table 15: Pairwise set judging against the controls (no graph and random slice) at high effort.

| Comparison | Pairs | First wins | Second wins | Ties | First-arm win rate | 95% CI | Sign test p | Order agreement |
|---|---|---|---|---|---|---|---|---|
| Full graph vs No graph | 65 | 44 | 13 | 8 | 74% | 62–83% | < 0.001 | 88% |
| Slice vs No graph | 65 | 34 | 19 | 12 | 62% | 49–72% | 0.053 | 82% |
| Slice vs Random graph | 65 | 25 | 20 | 20 | 54% | 42–65% | 0.55 | 68% |
| Full graph vs Slice (for reference) | 65 | 34 | 16 | 15 | 64% | 52–74% | 0.015 | 77% |

Mean score difference, first minus second arm (1–5 scale):

Table 16: Comparisons with the controls: mean difference in set scores per criterion, first minus second arm (1–5 scale).

| Comparison | Relevance | Novelty | Feasibility | Specificity | Diversity |
|---|---|---|---|---|---|
| Full graph − No graph | +0.21 | +0.24 | +0.29 | +0.88 | −0.11 |
| Slice − No graph | +0.16 | +0.20 | +0.14 | +0.52 | −0.30 |
| Slice − Random graph | +0.14 | −0.08 | +0.20 | −0.03 | −0.20 |

- **Full graph vs no graph:** a clear win for A on every criterion except diversity, above all specificity (+0.88). R vs C was not judged.
- **Need slice vs no graph:** B wins 62%, just short of significance ($p = 0.053$), again mainly through specificity. C's sets are judged the more diverse ($-0.30$): without a graph, the five ideas range more widely.
- **Need slice vs random slice:** no detectable difference. B's sets are slightly more relevant and feasible, R's slightly more diverse; 20 of 65 pairs are ties and the judge agreed with itself across orders for only 68% of pairs, the lowest of any comparison, so the two kinds of set are hard to tell apart.
- **Together with the tail:** the graph's set-level benefit grows with the amount of real product structure in the prompt (C < B < A). Relevance shows up mainly in the top tail (B 14.7% vs R 9.8% top ideas), not in the average set. This matters for the claim: a relevant slice helps find the best ideas, but the evidence that it beats a random slice of the same size comes from the tail measure alone.
- **Length does not explain the wins over no graph:** C's sets are longer than B's (529 vs 492 words), yet B wins; in A vs C the longer set won 68% of decided pairs, below A's 74%.

#### 4.4.4 Mean idea quality (individual scoring)

Table 17 gives the need-balanced mean idea score (1–5) and mean criterion scores.

Table 17: Mean idea score and mean criterion scores per arm (individual scoring, 1–5, balanced across market needs).

| Arm | Mean score | Relevance | Novelty | Feasibility | Specificity |
|---|---|---|---|---|---|
| No graph · high | 3.44 | 3.49 | 3.29 | 3.21 | 3.78 |
| Random graph · high | 3.46 | 3.44 | 3.16 | 3.37 | 3.88 |
| Slice · low | 3.28 | 3.38 | 3.01 | 3.26 | 3.50 |
| Slice · medium | 3.47 | 3.53 | 3.20 | 3.36 | 3.82 |
| Slice · high | 3.53 | 3.60 | 3.25 | 3.37 | 3.92 |
| Slice · xhigh (6 market needs) | 3.54 | 3.47 | 3.38 | 3.29 | 4.03 |
| Full graph · low | 3.30 | 3.39 | 3.04 | 3.27 | 3.52 |
| Full graph · medium | 3.44 | 3.42 | 3.10 | 3.42 | 3.83 |
| Full graph · high | 3.50 | 3.46 | 3.20 | 3.45 | 3.89 |
| Full graph · xhigh (6 market needs) | 3.52 | 3.42 | 3.33 | 3.29 | 4.03 |

- Mean differences between arms are small (0.1–0.25 points) and hide the differences in the tail; this is why the tail is the main measure.
- At high effort, mean scores order C 3.44 < R 3.46 < A 3.50 < B 3.53, the same ranking as the top-idea rates except that C and R, nearly tied on both measures, swap places.
- **No graph gives the most novel but least feasible single ideas** at high effort: novelty 3.29 (highest at high effort) and feasibility 3.21 (lowest). Without the graph, ideas are less constrained by the actual product. Judged as sets, though, A and B are rated more novel than C (Section 4.4.3): the individual novelty advantage of C does not survive a side-by-side comparison.
- **Individual ideas vs whole sets:** B's ideas score at least as well as A's individually, yet A wins 64% of set comparisons. A's advantage is at the level of the set (specificity, diversity), not in better single ideas.

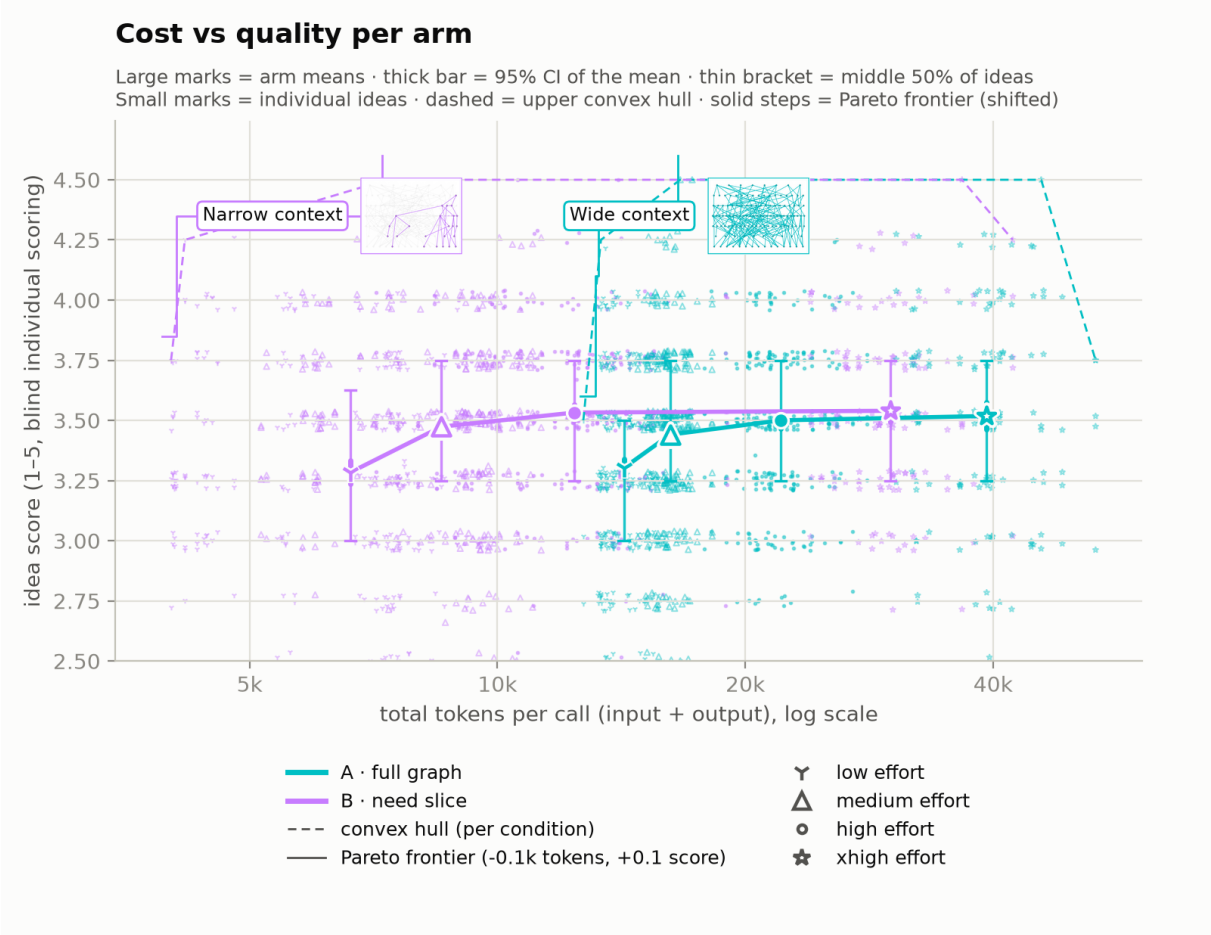



Figure 13: Tokens per call against mean idea score for the full-graph (A) and market-need-slice (B) arms at each effort. Moving right costs tokens; moving up raises the average idea, not necessarily the number of top ideas.

## 4.5 RQ3: How does spending tokens on context compare with spending them on thinking effort?

**Answer.** Thinking effort raises quality; the choice of context raises efficiency. High effort beats low in 92–94% of set comparisons and xhigh gives the highest top-idea rates (19–21%), but the cost per call rises in step, so yield does not improve: it is flat for the full graph and peaks at medium effort for the slice. For a similar budget per call, tokens are better spent on thinking with a relevant slice than on context: the slice at high effort beats the full graph at low effort in 91% of set comparisons, with 13% fewer tokens per call.

### 4.5.1 Effort, quality and yield

Figure 13 plots tokens per call against mean idea score for the full-graph and market-need-slice arms.

- **Effort:** top-idea rates rise with effort for both A and B, with the largest rates at xhigh. But xhigh costs 2–3× the tokens per top idea of medium or high.
- **The full graph's yield is flat across effort** (2.6–3.2). More thinking raises A's top-idea rate from 8% to 21%, but the cost per call rises in step, so the full graph buys better calls, not more top ideas per token.
- **The slice's yield peaks at medium effort.** From medium to high, B's top-idea rate

barely changes (15.1% to 14.7%) while the tokens per call rise by 45%; xhigh raises the rate to 18.7% but at 3.5 times the tokens.

#### 4.5.2 Pairwise set judging: effort levels (equal representation)

Table 18 gives the win rates between effort levels and Table 19 the differences per criterion.

Table 18: Pairwise set judging between effort levels within the same representation.

| Comparison | Condition | Pairs | Higher wins | Lower wins | Ties | Higher-effort win rate | 95% CI | Sign test p | Order agreement |
|---|---|---|---|---|---|---|---|---|---|
| high vs low | Full graph | 65 | 58 | 3 | 4 | 92% | 83–97% | $< 0.001$ | 94% |
| high vs low | Slice | 65 | 59 | 2 | 4 | 94% | 85–98% | $< 0.001$ | 94% |
| xhigh vs high | Full graph | 30 | 19 | 5 | 6 | 73% | 56–86% | 0.007 | 80% |
| xhigh vs high | Slice | 30 | 25 | 4 | 1 | 85% | 68–94% | $< 0.001$ | 97% |

Mean score difference, higher minus lower effort:

Table 19: Effort comparisons: mean difference in set scores per criterion, higher minus lower effort (1–5 scale).

| Comparison | Condition | Relevance | Novelty | Feasibility | Specificity | Diversity |
|---|---|---|---|---|---|---|
| high − low | Full graph | +0.48 | +0.71 | +0.52 | +1.14 | +0.63 |
| high − low | Slice | +0.71 | +0.69 | +0.70 | +1.23 | +0.71 |
| xhigh − high | Full graph | +0.25 | +0.34 | +0.02 | +0.46 | +0.33 |
| xhigh − high | Slice | +0.19 | +0.47 | +0.03 | +0.61 | +0.45 |

- High beats low on every criterion and on all 13 market needs.
- xhigh beats high again, mainly through specificity, novelty and diversity, not feasibility, at 1.8× (A) to 2.4× (B) the tokens. Effort had not saturated at the highest level tested.

#### 4.5.3 Equal budget: slice with high effort vs full graph with low effort

Table 20 compares two arms of similar cost per call.

Table 20: Market-need slice at high effort versus full graph at low effort, two arms of similar cost per call.

| Comparison | Pairs | Slice-high wins | Full-graph-low wins | Ties | Slice-high win rate | 95% CI | Sign test p | Order agreement |
|---|---|---|---|---|---|---|---|---|
| Slice high vs Full graph low | 65 | 56 | 3 | 6 | 91% | 81–96% | $< 0.001$ | 91% |

B-high was ahead on every criterion and for all 13 market needs, at 12,407 vs 14,243 tokens per call (13% fewer). For a similar budget, a relevant slice with more thinking clearly beats the full graph with little thinking.

### 4.6 Exploratory: persona as a factor

Does the best representation depend on who is thinking? A follow-up (`persona_as_factor/`, hypotheses H5–H7 written down before the run) crossed three personas with the market-need slice (B) and no graph (C), on three market needs (M4, M8, M13), at medium and high effort, 15 calls per cell (540 planned, 532 logged, 525 parsed and scored; the 8 missing calls did not complete on re-runs and are spread over five cells, 1–2 each, so they cannot change the picture). The innovation engineer is the main study's persona, verbatim; the 45 matching main-study calls were reused. The two new personas replace the first sentence of the system prompt with a description of a way of thinking and one credited exemplar:

- **Lateral thinker:** behavioural economics, changing perception rather than performance; exemplar Rory Sutherland's Eurostar observation (make the journey more enjoyable, e.g. with Wi-Fi, rather than the train faster).
- **Patent-focused inventor:** new, non-obvious, industrially applicable, precise enough to be claimed; exemplar James Dyson's bagless cyclone vacuum.

Ideas were scored with the main study's quality prompt, unchanged, and in a separate call for patentability (new, non-obvious, applicable, overall 1–5).

Table 21 gives the **yield (top ideas per 100k tokens), pooled over the three market needs and both efforts** (per effort: repository chart `10_persona_yield.png`; bootstrap spread per persona and representation: repository chart `11b_persona_by_representation.png`):

Table 21: Persona follow-up: yield (top ideas per 100k tokens, 95% bootstrap CI) per persona with the market-need slice (B) and with no graph (C), pooled over three market needs and two efforts.

| Persona | B · market-need slice | C · no graph | $B \div C$ [95% CI] | Top-idea rate B / C |
|---|---|---|---|---|
| Innovation engineer | 5.9 [4.8–7.2] | 7.0 [5.1–9.1] | 0.84 [0.60–1.22] | 13.4% / 9.3% |
| Lateral thinker | 4.6 [3.6–5.6] | 3.9 [2.7–5.2] | 1.16 [0.80–1.76] | 11.0% / 5.8% |
| Patent-focused | 5.8 [4.8–6.8] | 4.9 [3.6–6.4] | 1.17 [0.86–1.66] | 14.8% / 7.7% |

Table 22 gives **one yield per persona** (all its calls: both representations, both efforts; Figure 14, `11_persona_effect.png`):

Table 22: Persona follow-up: one yield per persona over all its calls, the chance it beats the innovation engineer, and mean patentability (1–5).

| Persona | Yield [95% CI] | P(beats engineer) | Mean patentability [95% CI] |
|---|---|---|---|
| Innovation engineer | 6.3 [5.3–7.4] | – | 2.19 [2.15–2.22] |
| Patent-focused | 5.5 [4.6–6.3] | 0.10 | 2.23 [2.19–2.27] |
| Lateral thinker | 4.3 [3.5–5.2] | $< 0.001$ | 2.17 [2.13–2.21] |

P = share of 2,000 paired bootstrap draws (calls resampled within market need) in which the persona's yield is higher than the engineer's. The patent persona's yield is higher than the lateral thinker's in 97% of draws.

- **No persona beat the plain engineer per token.** The engineer is the most efficient overall (the patent persona is behind but not clearly worse); the lateral thinker the least, most of all without a graph (5.8% top ideas, against 9.3% for the engineer). The new personas also cost slightly more per call: their descriptions add about 150 input tokens,

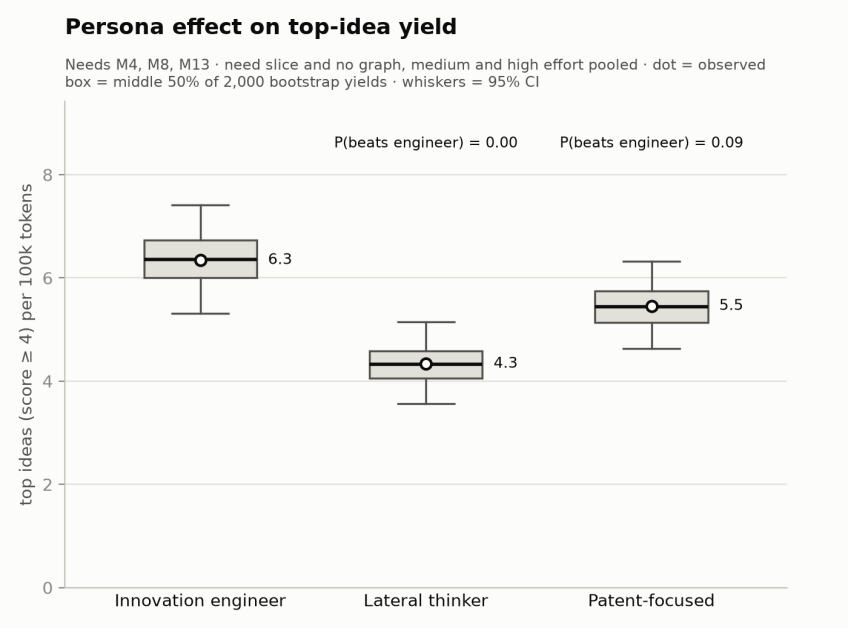



Figure 14: Persona follow-up: yield per persona over all of its calls (market-need slice and no graph, medium and high effort). Box = middle 50% of 2,000 bootstrap yields, whiskers = 95% CI, dot = observed.

and the patent persona thinks longer (about 11.7k output tokens per high-effort slice call, against 9.6k for the engineer).

- **The slice helped every persona's top-idea rate,** and the lateral thinker most: its rate nearly doubles with the slice (5.8% to 11.0%). A persona that works by reframing seems to need concrete product structure to anchor on.
- **The best single cell** is the patent persona with the slice at high effort: 18.0% top ideas, and the only $B \div C$ ratio whose interval clears 1 (1.55 [1.03–2.61]). It is one cell of twelve and is reported as a pattern.
- **On these three market needs, the engineer's slice does not beat no graph** ($B \div C$ 0.84 [0.60–1.22] pooled; at medium effort no graph is even ahead, 10.0 against 6.4, intervals overlapping). In the main study, too, no graph's yield at high effort (6.4) was close to the slice's (5.9); the slice's clear lead there came at medium effort on all 13 market needs (8.8), and on M4, one of the three market needs here, no graph did well. The slice's efficiency advantage over no graph is therefore market-need-dependent; its advantage over the full graph and the random slice is the robust result.
- **The exemplars were not copied:** at most 1.8% of ideas mention Wi-Fi, the Eurostar or entertainment (engineer baseline 0.7%), and none mention Dyson, cyclones or bagless designs.

**Against the hypotheses written down before the run:**

- **H5 (the lateral thinker needs the graph less): not supported.** The $B \div C$ ratio points the other way (1.16 against the engineer's 0.84), and the intervals overlap.
- **H6 (the patent persona needs the graph more, for patentable ideas): not testable.** The judge rated almost every idea 2 out of 5 for patentability (means 2.13–2.31 in every cell); only about 1% of ideas reach the pre-set threshold of 4, 0–2 per cell, so patentable yields are noise.
- **H7 (personas do what they are told): not supported.** The lateral thinker's ideas are not rated more novel (mean novelty 3.23, against 3.30–3.34 for the engineer), and

the patent persona's mean patentability is barely higher (2.23 [2.19–2.27] against 2.19 [2.15–2.22]; a difference of 0.04 on a 1–5 scale, intervals overlapping).

**Reading these results.** The quality rubric (relevance, novelty, feasibility, specificity) rewards concrete engineering, and the operator is SUBSTITUTE; a persona that proposes changes of perception rather than of parts may be marked down by the rubric rather than be worse. Patentability is the judge's estimate without a prior-art search; that it rates nearly all ideas as weakly patentable is consistent with many top ideas reusing known moves (solid-state RF, RF thawing), but it is not a legal assessment. Both are planned for the human check.

## 4.7 Robustness and caveats

### 4.7.1 Length and judge reliability

Table 23 checks whether the judge simply preferred longer sets.

Table 23: Length check: mean words per set in each comparison, how often the longer set won, and the winning arm's win rate.

| Comparison | Mean words per set (first / second arm) | Longer set won | Win rate of the winning arm |
|---|---|---|---|
| Full graph vs Slice, low effort | 452 / 443 | 65% | 65% (Full graph) |
| Full graph vs Slice, high effort | 542 / 492 | 71% | 64% (Full graph) |
| High vs low effort, Full graph | 542 / 452 | 90% | 92% (high) |
| High vs low effort, Slice | 492 / 443 | 85% | 94% (high) |
| xhigh vs high effort, Full graph | 587 / 530 | 88% | 73% (xhigh) |
| xhigh vs high effort, Slice | 587 / 499 | 83% | 85% (xhigh) |
| Slice high vs Full graph low | 492 / 452 | 74% | 91% (Slice high) |
| Full graph vs No graph, high effort | 542 / 529 | 68% | 74% (Full graph) |
| Slice vs No graph, high effort | 492 / 529 | 60% | 62% (Slice) |
| Slice vs Random graph, high effort | 492 / 492 | 59% | 54% (Slice) |

- **A vs B at low effort:** sets are nearly the same length, so A's set-level advantage is not explained by length.
- **Effort:** each step up produces longer sets, so part of the effort effect may be detail bias in the judge.
- **B-high vs A-low:** B-high wins 91% but the longer set wins only 74%, so B-high also wins pairs where its ideas are shorter.
- **B vs C:** B wins although its sets are shorter, so the graph's advantage over no graph is not a length effect.
- **Order agreement:** 68–97% of pairs got the same verdict in both orders (85% overall); split verdicts count as ties. The lowest is B vs R (68%), where the sets are hardest to tell apart.
- **Individual scores** do not depend on length comparisons between arms, which makes the tail measure less exposed to detail bias than the pairwise judging.

### 4.7.2 Change from the pilot

The pilot (2 market needs × 3 replicates) found the full graph winning 17 of 18 set comparisons. At full scale (13 market needs, 65 pairs) A's set-level advantage shrinks to about 64%, and on the tail measure it disappears. The pilot's "more novel but less feasible" pattern for A did not replicate (it now describes no graph, C). The effort effect and the B-high vs A-low result held. Contrary to the pilot's expectation, quality kept rising from high to xhigh.

#### 4.7.3 Caveats

- **Single judge model,** with no human calibration yet; a "top idea" is one the judge rates highly on relevance, novelty, feasibility and specificity together, a grounded strong idea rather than necessarily the wildest one.
- **Coarse scores:** individual scores move in 0.25 steps, so the threshold (4.0, top 12.3%) is the nearest available cut, and the 4.25 tail has too few ideas to separate arms.
- **Yield assumes a constant rate per token.** The observed running counts are close to straight lines (Figure 7), but they depend on the order in which calls are taken; the slope itself does not.
- **Budget curves assume independent calls** within a market need and use a small-sample estimate of each market need's success rate; they are a model, not observed sequences.
- **R and C were run only at high effort;** R vs C was not judged pairwise, so whether a random slice beats no graph on whole sets is unknown.
- **Relevance vs size rests on the tail measure:** the pairwise judging found no difference between B and R.
- **R controls for node count, not edges or graph distribution:** $R$ matches $B$ in node count and layer composition, but random selection yields substantially sparser induced graphs (2–52 edges vs. 11–85 in $B$) and 17% fewer input tokens. Furthermore, only a single fixed random draw per market need was evaluated rather than an ensemble over multiple random graph realizations, so variance across different random subgraphs is not averaged out.
- **Length and effort are intertwined;** the pairwise effort effects may be partly inflated by detail bias.
- **xhigh covers 6 of 13 market needs,** which are slightly easier than average for top ideas; see the matched-needs results.
- **One product, one operator, one prompt.** The worked example comes from the M8 branch of the graph, which may favour ideas in that area.
- **Per-market-need results** rest on 5 calls or pairs each and should be read as patterns; the slice-size effect is exploratory (13 market needs, $p \approx 0.04$–$0.13$).

# 5 Discussion and conclusion

## 5.1 Conclusions

This study asked how much of a product's knowledge graph an LLM needs to see in order to brainstorm well, measured not by the quality of its output alone but by how efficiently it turns tokens into good ideas. An innovation matrix (Tchirsky, n.d.) of a microwave oven (123 nodes, 246 edges) was given to `claude-haiku-5-5` in four forms: the full graph (A), the vertical slice for the target market need (B), a random slice of the same size (R), or no graph at all (C). The model applied (individually to each market need) the SUBSTITUTE operator to 13 market needs, with 5 replicates per cell and four levels of thinking effort. The 2,892 resulting ideas were judged blind by `claude-sonnet-5-5`, both as whole sets side by side and one idea at a time.

Brainstorming is a search for the rare great idea, so the main measure was the **yield**: top ideas (the top ∼12% of all ideas) found per 100k tokens spent. Counted against tokens, top ideas accumulate in a straight line for every arm, so arms can be compared by slope. Five conclusions follow.

1. **A relevant slice of the graph finds top ideas fastest.** The market-need slice at medium effort yields 8.8 top ideas per 100k tokens, about three times the full graph at any effort (2.6–3.2). The advantage over the full graph holds market need by market need: the slice is more efficient on 10 of 13 market needs (sign test $p = 0.039$), and its lead looks largest where the slice is small (exploratory).

2. **More context buys better average sets, not more great ideas.** Judged as whole sets, the full graph beats the slice (64%) and no graph (74%), mainly by being more specific. But its top-idea rate is no higher than the slice's, and its calls cost about twice as much, so per token it finds fewer top ideas than giving the model no graph at all (3.1 against 6.4 at high effort).

3. **Relevance matters, size does not.** A random slice of the same size as the market-need slice is less efficient than both the market-need slice and no graph, though still ahead of the full graph: the market-need slice's slope is higher in 99% of paired bootstrap draws (5.9 against 4.1 at high effort), and no graph beats the random slice on 10 of 12 market needs ($p = 0.039$). Irrelevant structure is worse per token than none. Judged as whole sets, market-need slice and random slice cannot be told apart: relevance shows up in the best ideas, not in the average set.

4. **Thinking effort raises quality, not efficiency.** High effort beats low in about 93% of set comparisons, and extra-high effort gives the highest top-idea rates (19–21%). But each call costs proportionally more, so yield does not rise with effort: the full graph stays at about 3 top ideas per 100k tokens from low to extra-high, and the slice peaks at medium effort.

5. **A persona prompt is no substitute for the right context (exploratory).** In a follow-up on three market needs, neither a lateral-thinker persona nor a patent-focused persona beat the plain innovation engineer per token (6.3 top ideas per 100k tokens against 5.5 and 4.3), and the hypotheses that they would change which representation works best were not supported. The lateral thinker was the least efficient, especially without a graph; given the market-need slice, its top-idea rate nearly doubled. The judge rated almost no idea from any persona as clearly patentable.

In summary, Haiku is a small model, what such a model does superbly is retrieval: it holds a lossy copy of its training corpus, patents and product literature included, and surfaces the engineering moves that bear on a stated problem (Section 3.10). A retrieval system is most efficient when the query is precise.

### 5.2 Implications for design thinking

- **Framing beats tokens.** The market-need slice is a machine version of a familiar design move: narrowing the problem.
- **Yield turns ideation into a budgetable activity.**
- **Context behaved like an input with diminishing, then negative, returns.**
- **Representation mattered more than role.**
- **The product graph is a reusable, auditable design asset.**

### 5.3 New research directions

- **Graph-guided search instead of prompts.**
- **Operators as graph transformations.** TRIZ (40) principles and SCAMPER (Eberle, 1971) can each be defined as operations on the graph.
- **Yield as a standard measure.** Top ideas per token is simple, transferable and does not saturate.

## References


Yoji Akao, editor. *Quality Function Deployment: Integrating Customer Requirements into Product Design*. Productivity Press, New York, 2024. doi: 10.4324/9781003578833. Reissue of the 1990 edition.

Chenxin An, Jun Zhang, Ming Zhong, Lei Li, Shansan Gong, Yao Luo, Jingjing Xu, and Lingpeng Kong. Why Does the Effective Context Length of LLMs Fall Short? arXiv preprint arXiv:2410.18745, 2024.

Barrett R Anderson, Jash Hemant Shah, and Max Kreminski. Homogenization Effects of Large Language Models on Human Creative Ideation. *Creativity and Cognition 2024*, 2024. doi: 10.1145/3635636.3656204.

Joshua Ashkinaze, Julia Mendelsohn, Li Qiwei, Ceren Budak, and Eric Gilbert. How AI Ideas Affect the Creativity, Diversity, and Evolution of Human Ideas: Evidence From a Large, Dynamic Experiment. *Proceedings of the ACM Collective Intelligence Conference 2025*, 2025. doi: 10.1145/3715928.3737481.

Jose Berengueres. *The Brown Book of Design Thinking*. UAE University College, Al Ain, 2014.

T. R. Browning. Applying the design structure matrix to system decomposition and integration problems: a review and new directions. *IEEE Transactions on Engineering Management 48(3)*, 2001. doi: 10.1109/17.946528.

Lai-Kow Chan and Ming-Lu Wu. Quality function deployment: A literature review. *European Journal of Operational Research 143(3)*, 2002. doi: 10.1016/s0377-2217(02)00178-9.

Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. *Transactions on Machine Learning Research*, 2024a.

Liuqing Chen, Qianzhi Jing, Yixin Tsang, Qianyi Wang, Lingyun Sun, and Jianxi Luo. DesignFusion: Integrating Generative Models for Conceptual Design Enrichment. *Journal of Mechanical Design*, 2024b. doi: 10.1115/1.4065487.

Liuqing Chen, Yaxuan Song, Shixian Ding, Lingyun Sun, Peter Childs, and Haoyu Zuo. TRIZ-GPT: An LLM-Augmented Method For Problem-Solving. *ASME IDETC-CIE 2024*, 2024c. doi: 10.1115/detc2024-143163.

Liuqing Chen, Yiyan Tsang, Qianzhi Jing, and Lingyun Sun. A LLM-augmented Morphological Analysis Approach for Conceptual Design. *Proceedings of DRS2024*, 2024d. doi: 10.21606/drs.2024.605.

Liuqing Chen, Haoyu Zuo, Zebin Cai, Yuan Yin, Yuan Zhang, Lingyun Sun, Peter Childs, and Boheng Wang. Toward Controllable Generative Design: A Conceptual Design Generation Approach Leveraging the Function–Behavior–Structure Ontology and Large Language Models. *Journal of Mechanical Design*, 2024e. doi: 10.1115/1.4065562.

Ely Dahan and Haim Mendelson. An Extreme-Value Model of Concept Testing. *Management Science*, 47(1):102–116, 2001. doi: 10.1287/mnsc.47.1.102.10666.

Edward de Bono. *Lateral Thinking: Creativity Step by Step*. Harper & Row, New York, 1970.

Dujian Ding, Ankur Mallick, Chi Wang, et al. Hybrid LLM: Cost-Efficient and Quality-Aware Query Routing. arXiv preprint arXiv:2404.14618, 2024.

Shiying Ding, Xinyi Chen, Yan Fang, Wenrui Liu, Yiwu Qiu, and Chunlei Chai. DesignGPT: Multi-Agent Collaboration in Design. *2023 16th International Symposium on Computational Intelligence and Design (ISCID)*, 2023. doi: 10.1109/iscid59865.2023.00056.

Anil R. Doshi and Oliver P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 2024. doi: 10.1126/sciadv.adn5290.

Bob Eberle. *SCAMPER: Games for Imagination Development*. D.O.K. Publishers, Buffalo, NY, 1971.

Steven D. Eppinger and Tyson R. Browning. *Design Structure Matrix Methods and Applications*. MIT Press, Cambridge, MA, 2012. doi: 10.7551/mitpress/8896.001.0001.

Lee Fleming. Recombinant Uncertainty in Technological Search. *Management Science 47(1)*, 2001. doi: 10.1287/mnsc.47.1.117.10671.

J. S. Gero. Design Prototypes: A Knowledge Representation Schema for Design. *AI Magazine 11(4)*, 1990. doi: 10.1609/aimag.v11i4.854.

John S. Gero and Udo Kannengiesser. The situated function–behaviour–structure framework. *Design Studies 25(4)*, 2004. doi: 10.1016/j.destud.2003.10.010.

Karan Girotra, Christian Terwiesch, and Karl T. Ulrich. Idea Generation and the Quality of the Best Idea. *Management Science*, 56(4):591–605, 2010. doi: 10.1287/mnsc.1090.1144.

Tingxu Han, Zhenting Wang, Chunrong Fang, et al. Token-Budget-Aware LLM Reasoning. *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. doi: 10.18653/v1/2025.findings-acl.1274.

John R. Hauser and Don Clausing. The House of Quality. *Harvard Business Review*, 66(3): 63–73, 1988.

Julie Hirtz, Robert B. Stone, Daniel A. McAdams, Simon Szykman, and Kristin L. Wood. A functional basis for engineering design: Reconciling and evolving previous efforts. *Research in Engineering Design 13(2)*, 2002. doi: 10.1007/s00163-001-0008-3.

Shuo Jiang, Weifeng Li, Yuping Qian, Yangjun Zhang, and Jianxi Luo. AutoTRIZ: Automating engineering innovation with TRIZ and large language models. *Advanced Engineering Informatics*, 2025. doi: 10.1016/j.aei.2025.103312.

Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, et al. AI Agents That Matter. *Transactions on Machine Learning Research*, 2025.

Tom Kelley and Jonathan Littman. *The Ten Faces of Innovation: IDEO's Strategies for Defeating the Devil's Advocate and Driving Creativity Throughout Your Organization*. Currency/Doubleday, New York, 2005.

Steven Kyffin and Paul Gardien. Navigating the Innovation Matrix: An Approach to Design-led Innovation. *International Journal of Design*, 3(1):57–69, 2009.

Mosh Levy, Alon Jacoby, and Yoav Goldberg. Same Task, More Tokens: the Impact of Input Length on the Reasoning Performance of Large Language Models. *Proceedings of ACL 2024 (Long Papers)*, 2024. doi: 10.18653/v1/2024.acl-long.818.

Zhuowan Li, Cheng Li, Mingyang Zhang, et al. Retrieval Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach. *Proceedings of EMNLP 2024: Industry Track*, 2024. doi: 10.18653/v1/2024.emnlp-industry.66.

Beibei Liu et al. Creative as IDEO Experts: LLM-Agent-Based Design Thinking Workshop. In *Proceedings of the 16th International Conference on Computational Creativity (ICCC'25)*, 2025.

Nelson F. Liu, Kevin Lin, John Hewitt, et al. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 2024. doi: 10.1162/tacl_a_00638.

Songkai Liu, Yanqing Shen, Yilun Zhang, Zhangli Hou, Xin Wang, Jianxi Luo, and Zhinan Zhang. iDesignGPT enhances conceptual design via large language model agentic workflows. *Nature Communications*, 2026. doi: 10.1038/s41467-026-68672-1.

Li-Chun Lu, Shou-Jen Chen, Tsung-Min Pai, Chan-Hung Yu, Hung yi Lee, and Shao-Hua Sun. LLM Discussion: Enhancing the Creativity of Large Language Models via Discussion Framework and Role-Play. arXiv preprint arXiv:2405.06373, 2024.

Jianxi Luo, Serhad Sarica, and Kristin L. Wood. Guiding data-driven design ideation by knowledge distance. *Knowledge-Based Systems 218*, 2021. doi: 10.1016/j.knosys.2021.106873.

Kevin Ma, Daniele Grandi, Christopher McComb, and Kosa Goucher-Lambert. Conceptual Design Generation Using Large Language Models. *ASME IDETC-CIE 2023*, 2023. doi: 10.1115/detc2023-116838.

Kevin Ma, Daniele Grandi, Christopher McComb, and Kosa Goucher-Lambert. Do Large Language Models Produce Diverse Design Concepts? A Comparative Study with Human-Crowdsourced Solutions. *Journal of Computing and Information Science in Engineering*, 2025. doi: 10.1115/1.4067332.

Soheyl Massoudi and Mark Fuge. Agentic Large Language Models for Conceptual Systems Engineering and Design. *Journal of Mechanical Design*, 2026. doi: 10.1115/1.4070328.

Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, et al. NoLiMa: Long-Context Evaluation Beyond Literal Matching. arXiv preprint arXiv:2502.05167, 2025.

Isaac Ong, Amjad Almahairi, Vincent Wu, et al. RouteLLM: Learning to Route LLMs with Preference Data. arXiv preprint arXiv:2406.18665, 2024.

Alex F. Osborn. *Applied Imagination: Principles and Procedures of Creative Problem-Solving.* Charles Scribner's Sons, New York, 1953.

Norman Paulsen. Context Is What You Need: The Maximum Effective Context Window for Real World Limits of LLMs. *Advances in Artificial Intelligence and Machine Learning*, 2026. doi: 10.54364/AAIML.2026.61268.

Robert Phaal, Clare J. P. Farrukh, and David R. Probert. Technology roadmapping—A planning framework for evolution and revolution. *Technological Forecasting and Social Change 71(1–2)*, 2004. doi: 10.1016/s0040-1625(03)00072-6.

T. Ritchey. Problem structuring using computer-aided morphological analysis. *Journal of the Operational Research Society 57(7)*, 2006. doi: 10.1057/palgrave.jors.2602177.

Tom Ritchey. General morphological analysis as a basic scientific modelling method. *Technological Forecasting and Social Change 126*, 2018. doi: 10.1016/j.techfore.2017.05.027.

Serhad Sarica, Jianxi Luo, and Kristin L. Wood. TechNet: Technology semantic network based on patent data. *Expert Systems with Applications 142*, 2020. doi: 10.1016/j.eswa.2019.112995.

Prabir Sarkar and Amaresh Chakrabarti. Assessing design creativity. *Design Studies 32(4)*, 2011. doi: 10.1016/j.destud.2011.01.002.

Jami J. Shah, Steve M. Smith, and Noe Vargas-Hernandez. Metrics for measuring ideation effectiveness. *Design Studies*, 2003. doi: 10.1016/s0142-694x(02)00034-0.

Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers. arXiv preprint arXiv:2409.04109, 2024.

Donald V. Steward. The design structure system: A method for managing the design of complex systems. *IEEE Transactions on Engineering Management EM-28(3)*, 1981. doi: 10.1109/tem.1981.6448589.

Robert B. Stone and Kristin L. Wood. Development of a Functional Basis for Design. *Journal of Mechanical Design 122(4)*, 2000. doi: 10.1115/1.1289637.

Kamil Szczepanik and Jarosław Chudziak. TRIZ Agents: A Multi-Agent LLM Approach for TRIZ-Based Innovation. *Proceedings of ICAART 2025 (SciTePress)*, 2025. doi: 10.5220/0013321900003890.

Leo Tchirsky. [Innovation matrix: reference to be added], n.d.

Christian Terwiesch, Lennart Meincke, Karan Girotra, Ethan Mollick, Gideon Nave, and Karl Ulrich. Artificial intelligence and its impact on creativity and diversity: An empirical study of large language model-generated product ideas. *Production and Operations Management*, 2026. doi: 10.1177/10591478261474243.

Brian Uzzi, Satyam Mukherjee, Michael Stringer, and Ben Jones. Atypical Combinations and Scientific Impact. *Science 342(6157)*, 2013. doi: 10.1126/science.1240474.

Dennis Verhoeven, Jurriën Bakker, and Reinhilde Veugelers. Measuring technological novelty with patent-based indicators. *Research Policy 45(3)*, 2016. doi: 10.1016/j.respol.2015.11.010.

Samangi Wadinambiarachchi, Ryan M. Kelly, Saumya Pareek, Qiushi Zhou, and Eduardo Velloso. The Effects of Generative AI on Design Fixation and Divergent Thinking. *Proceedings of CHI 2024*, 2024. doi: 10.1145/3613904.3642919.

Junlin Wang, Siddhartha Jain, Dejiao Zhang, et al. Reasoning in Token Economies: Budget-Aware Evaluation of LLM Reasoning Strategies. *Proceedings of EMNLP 2024*, 2024. doi: 10.18653/v1/2024.emnlp-main.1112.

Ningning Wang, Xavier Hu, Pai Liu, et al. Efficient Agents: Building Effective Agents While Reducing Cost. arXiv preprint arXiv:2508.02694, 2025.

Qihao Zhu and Jianxi Luo. Generative Transformers for Design Concept Generation. *Journal of Computing and Information Science in Engineering*, 2022. doi: 10.1115/1.4056220.

F. Zwicky. The Morphological Approach to Discovery, Invention, Research and Construction. In *New Methods of Thought and Procedure*, pages 273–297. Springer, 1967. doi: 10.1007/978-3-642-87617-2_14.

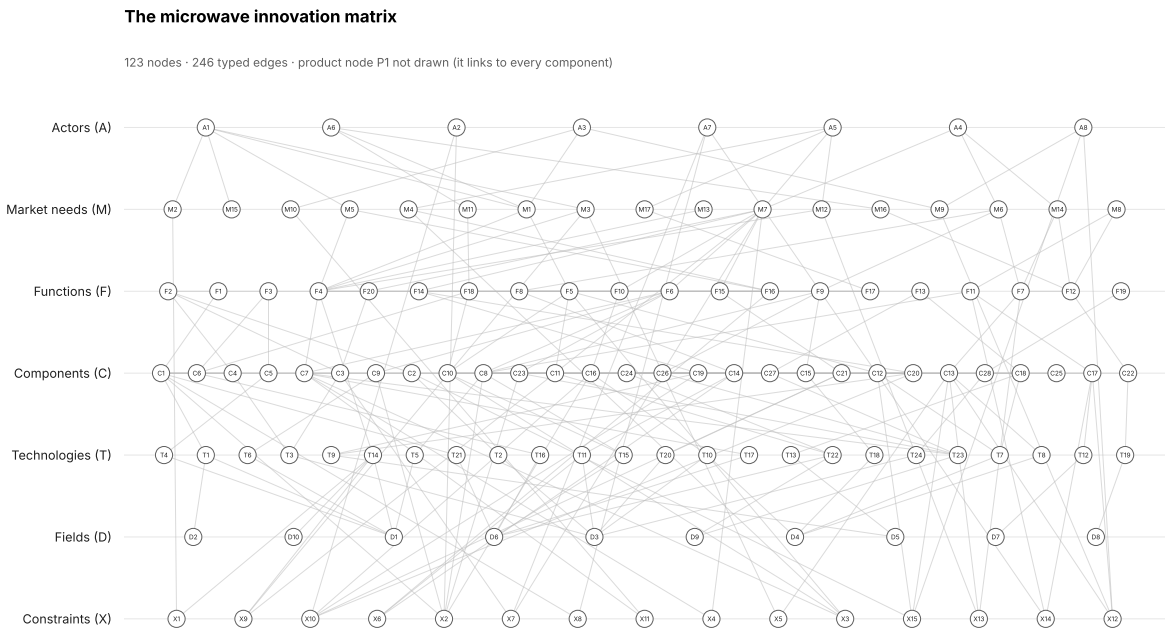



Figure 15: The full microwave innovation matrix (condition A).

# A The microwave innovation matrix

Figure 15 shows the whole graph. All 123 nodes are listed below by layer; the 246 edges and their types are in `microwave_base_graph.yaml`.

**Market needs (17)**

| ID | Label |
|---|---|
| M1 | Fast reheating for busy households |
| M2 | Crispy, browned texture like conventional oven |
| M3 | Hot milk / drink ready for breakfast |
| M4 | Defrost frozen food quickly |
| M5 | Save time in the kitchen |
| M6 | Simple operation with no learning curve |
| M7 | Safe for family use (radiation, burns, electric shock) |
| M8 | Human need for feedback (know when food is ready) |
| M9 | Low purchase price |
| M10 | Small kitchen footprint |
| M11 | Easy to clean |
| M12 | Healthy diet (nutrient retention, less fat) |
| M13 | Energy efficiency |
| M14 | Accessible for elderly / visually impaired users |
| M15 | Appliances that make you feel better (latent need) |
| M16 | Need for mobility (not having to stand by the appliance) |
| M17 | Bake a birthday cake (prep tasks like melting butter) |

**Functions (20)**

| ID | Label |
|---|---|
| F1 | Generates microwave energy at 2.45 GHz |
| F2 | Converts mains voltage to ∼4 kV DC for the magnetron |
| F3 | Guides microwave energy into the cavity |
| F4 | Heats food via dielectric water-molecule vibration (FastHeating) |

| ID | Label |
|---|---|
| F5 | Distributes heat evenly by rotating food through the field |
| F6 | Contains microwave radiation inside the appliance |
| F7 | Sets and counts down cooking duration |
| F8 | Sets average heating power level |
| F9 | Opens and closes the door to access the cavity |
| F10 | Cuts RF power instantly when the door opens |
| F11 | Illuminates cavity so the user can watch the food |
| F12 | Signals end of cooking (audible alert) |
| F13 | Cools magnetron and electronics |
| F14 | Protects against overheating |
| F15 | Protects against overcurrent and electrical faults |
| F16 | Defrosts frozen food (low-power cycling) |
| F17 | Melts / softens butter, chocolate, etc. |
| F18 | Contains spills and allows cleaning |
| F19 | Suppresses electrical noise conducted back to mains |
| F20 | Discharges stored high voltage after switch-off |

**Components (28)**

| ID | Label |
|---|---|
| C1 | Magnetron tube (2.45 GHz, ∼700 W output) |
| C2 | High-voltage transformer |
| C3 | High-voltage capacitor (∼1 $\mu$F, 2.1 kV) with internal bleeder resistor |
| C4 | High-voltage rectifier diode |
| C5 | Waveguide duct |
| C6 | Mica waveguide cover |
| C7 | Enamelled steel cooking cavity (∼20 L) |
| C8 | Hinged door with perforated metal mesh window |
| C9 | Door choke seal (quarter-wave rim) |
| C10 | Rotating glass turntable tray |
| C11 | Roller ring and coupler (turntable support) |
| C12 | Turntable motor (∼5 rpm) |
| C13 | Mechanical timer knob & cam switch (0-30 min) |
| C14 | Power-level knob (5-step duty-cycle cam switch) |
| C15 | Door-release push button & latch lever |
| C16 | Door interlock microswitches (primary, secondary, monitor) |
| C17 | Interior lamp (20-25 W incandescent) |
| C18 | Cooling fan and fan motor |
| C19 | Magnetron thermal cutout |
| C20 | Cavity thermal cutout |
| C21 | Main ceramic fuse |
| C22 | End-of-cycle bell (mechanical ‘ding’) |
| C23 | Outer steel cabinet (wrapper) |
| C24 | Control panel fascia |
| C25 | Mains power cord and plug |
| C26 | Door hinges |
| C27 | Ventilation louvres |
| C28 | Mains EMI noise filter |

**Technologies (24)**

| ID | Label |
|---|---|
| T1 | Electron cyclotron motion in crossed $E \times B$ fields (cavity magnetron) |
| T2 | Magnetic-shunt leakage HV transformer |
| T3 | Half-wave voltage doubler (capacitor + diode) |
| T4 | Rectangular waveguide TE10 mode propagation |

| ID | Label |
|---|---|
| T5 | Sub-wavelength aperture shielding (Faraday cage effect) |
| T6 | Quarter-wave choke seal |
| T7 | Shaded-pole / synchronous AC micro-motor (electromagnetic induction) |
| T8 | Motor-driven cam timer with spring-return mechanism |
| T9 | Bimetallic snap-disc thermostat |
| T10 | Redundant interlock microswitch logic (primary / secondary / monitor) |
| T11 | Duty-cycle power modulation (on/off cycling) |
| T12 | Incandescent tungsten-filament lamp |
| T13 | Forced-air convection cooling |
| T14 | Dielectric heating (dipolar rotation of water molecules) |
| T15 | Tempered / borosilicate glass forming |
| T16 | Mica sheet dielectric (muscovite laminate) |
| T17 | Enamel / epoxy powder coating of steel |
| T18 | Spring-loaded lever latch mechanics |
| T19 | Mechanical bell striker |
| T20 | Ceramic cartridge fuse (fusible-link overcurrent protection) |
| T21 | Sintered ferrite permanent magnets |
| T22 | Injection-molded engineering thermoplastics (PP, PBT, PPS) |
| T23 | Sheet-steel stamping & spot welding |
| T24 | Passive LC EMI line filtering |

**Fields (10)**

| ID | Label |
|---|---|
| D1 | Radio-frequency & microwave engineering |
| D2 | Vacuum electronics |
| D3 | Power electronics & electrical engineering |
| D4 | Electromechanics & mechanisms |
| D5 | Thermal engineering & heat transfer |
| D6 | Materials science (metals, ceramics, glass, polymers) |
| D7 | Lighting & optics |
| D8 | Sound engineering |
| D9 | Manufacturing engineering (stamping, molding, assembly) |
| D10 | Food science (dielectric properties of food) |

**Constraints (15)**

| ID | Label |
|---|---|
| X1 | Microwaves can't brown or crisp food surfaces |
| X2 | FCC/FDA radiation-leakage safety limits (21 CFR 1030.10, 5 mW/cm$^2$ at 5 cm) |
| X3 | Uneven heating — standing-wave hot and cold spots |
| X4 | Metal objects in cavity cause arcing |
| X5 | Mechanical timer imprecise for short (<1 min) durations |
| X6 | Power only adjustable by on/off cycling, not continuously |
| X7 | HV capacitor can hold a lethal charge after unplugging |
| X8 | Limited magnetron lifetime (cathode/filament wear) |
| X9 | Superheating of liquids — risk of eruptive boiling |
| X10 | Electrical safety standards (IEC 60335-2-25 / UL 923) |
| X11 | Cavity volume limited by countertop size |
| X12 | Low BOM target for entry-level price segment |
| X13 | No food-state sensing — open-loop cooking |
| X14 | Interior lamp burns out and is hard to replace |
| X15 | Regional mains variants (230 V/50 Hz vs 120 V/60 Hz) |

**Actors (8)**

| ID | Label |
|---|---|
| A1 | Time-pressed home cook |
| A2 | Appliance safety regulator/inspector |
| A3 | Student / single-person household |
| A4 | Elderly user |
| A5 | Parent with young children |
| A6 | Office worker using a shared kitchen |
| A7 | Repair technician |
| A8 | OEM manufacturer / cost engineer |

**Product (1)**

| ID | Label |
|---|---|
| P1 | Basic countertop microwave oven (timer knob, power knob, door button, lamp) |

# B Generation prompt

The template used for every call (`template_prompt.txt`). Placeholders in braces are filled per call; `{persona}` is "You are an innovation engineer." in the main study.

```
=== SYSTEM ===
{persona} You read product graphs and propose radical
product innovations. Respond with valid JSON only, no prose outside the JSON.

=== USER ===
Innovate the following {product_name} based on the graph given.
Encourage radical and wild ideas.

TARGET MARKET NEED
{target_need_id}: {target_need_label}
Favour innovations that serve this market need.

INNOVATION OPERATOR: {operator_name}
{operator_definition}

EXAMPLE OF AN INNOVATION WE LIKE
A microwave has a door with holes, a light, and other components and
functions, so the user can see inside: they need to check for mistakes
(perhaps there is also entertainment value).
Applying the operator: replace the light and the holes with something
that fulfils the same function equally well or better, or more cheaply.
  - e.g. 1: a webcam
  - e.g. 2: a fully transparent microwave casing that needs no light bulb inside

GRAPH OF THE {product_name}
{graph_yaml}

OUTPUT
Return exactly {n_ideas} ideas as JSON:
{
  "ideas": [
    {
      "title": "short name",
      "anchors": ["existing node ids the operator was applied to"],
      "new_nodes": [
        {"type": "Field|Technology|Component|Function|MarketNeed|Constraint",
```

```
        "label": "...", "detail": "one line"}
      ],
      "new_edges": [
        {"from": "node id or new node label", "to": "node id or new node label",
         "type": "one of the graph's edge_types, e.g. resolves"}
      ],
      "rationale": "one sentence: how this serves {target_need_id}"
    }
  ]
}
```

# C Pairwise judge prompt

System prompt:

```
You are an expert panel judging product-innovation brainstorming. You compare two
↪  sets of ideas produced for the same brief. Judge only the ideas themselves;
↪  ignore length and formatting. Respond with valid JSON only.
```

User prompt:

```
BRIEF
Product: {product}
Target market need: {need_id}: {need_label}
Innovation operator: {operator}
The brief asked for radical, wild ideas that serve the target need.

SET 1
{set1}

SET 2
{set2}

Score each set from 1 (poor) to 5 (excellent) on:
- relevance: how well the ideas serve the target need
- novelty: how radical and non-obvious the ideas are
- feasibility: how technically plausible the ideas are
- specificity: how concrete the ideas are (named components, mechanisms)
- diversity: how different the ideas are from each other

Then pick the better set overall for this brief.

Return JSON:
{"set_1": {"relevance": 0, "novelty": 0, "feasibility": 0, "specificity": 0,
↪  "diversity": 0},
 "set_2": {"relevance": 0, "novelty": 0, "feasibility": 0, "specificity": 0,
 ↪  "diversity": 0},
 "winner": "set_1" | "set_2" | "tie",
 "reason": "one sentence"}
```

# D Individual scoring prompts

Quality (all ideas). System prompt:

```
You are an expert panel judging product-innovation ideas. Rate each idea on its own
↪  merits, independently of the others. Ignore writing style and length. Respond
↪  with valid JSON only.
```

User prompt:

```
BRIEF
Product: {product}
Target market need: {need_id}: {need_label}
Innovation operator: {operator}
The brief asked for radical, wild ideas that serve the target need.

IDEAS
{ideas}

Rate EACH idea from 1 (poor) to 5 (excellent) on:
- relevance: how well it serves the target need
- novelty: how radical and non-obvious it is
- feasibility: how technically plausible it is
- specificity: how concrete it is (named components, mechanisms)

Return JSON with one entry per idea, in the same order:
{"ratings": [{"idea": 1, "relevance": 0, "novelty": 0, "feasibility": 0,
↪  "specificity": 0,
              "comment": "one sentence"}]}
```

Patentability (persona follow-up only), a separate call. System prompt:

```
You are an experienced patent examiner assessing product-innovation ideas. Rate each
↪  idea on its own merits, independently of the others. Ignore writing style and
↪  length. Respond with valid JSON only.
```

User prompt:

```
BRIEF
Product: {product}
Target market need: {need_id}: {need_label}
Innovation operator: {operator}
The brief asked for radical, wild ideas that serve the target need.

IDEAS
{ideas}

Rate EACH idea from 1 (poor) to 5 (excellent) on the three requirements for a patent:
- new: not already known, on the market or published in the form described
- non_obvious: not an obvious step for a skilled engineer in the field
- applicable: can be made and used as described (industrial applicability)
and give an overall patentability from 1 to 5: how likely a patent with a meaningful
↪  claim
could be granted for the idea as described. You cannot search prior art; use your
↪  knowledge
of existing products, patents and published work.

Return JSON with one entry per idea, in the same order:
{"ratings": [{"idea": 1, "new": 0, "non_obvious": 0, "applicable": 0,
↪  "patentability": 0,
              "comment": "one sentence"}]}
```

# E Data and code

| File | Contents |
|---|---|
| `runs/rq1_full.jsonl` | all 580 generation calls |
| `results/calls.csv` | one row per call: tokens, length and structure measures |
| `results/summary_condition_effort.csv`, `results/summary_cells.csv` | cost and validity, per arm and per market need |
| `results/judgments.jsonl`, `results/pairwise_summary.csv` | pairwise set verdicts; win rates, confidence intervals, sign tests, length check |
| `results/idea_scores.jsonl`, `results/ideas_ranked.csv`, `results/top_ideas.md` | individual idea scores, ranked |
| `persona_as_factor/` | persona follow-up: design and hypotheses (`README.md`), runs, scores, `results/persona_summary.md`, supplementary chart 10, 11, 11b |
| `yield_per_market_need.xlsx` | per-market-need top ideas and tokens per arm, yields, $B \div A$ ratios and market need-level sign tests (live formulas) |
| `results/tail_summary.csv` (+ `_ge4.25`, `_matched`) | top-idea rates, yields (top ideas per 100k tokens), confidence intervals, tokens per top idea |
| `results/charts/` | 0 the innovation matrix (full, and with the M8 slice; `figure_innovation_matrix.py`) · 1 tokens per call · 2 win rates · 3 score differences · 4 cost vs quality · 5 context ladder · 6 top-idea rate · 7 chance of a top idea vs budget · 8 top ideas vs tokens spent (yield) · 8b/8c spread of the slopes (8c annotated) · 9 slice-size effect per market need |